

INSTITUT UNIVERSITAIRE DES SCIENCES

FALCUTÉ DES SCIENCES ET TECHNOLOGIE (FST)

RAPPORT DE TRAVAUX DIRIGÉS I : INTRODUCTION À CISCO PACKET TRACER ET TOPOLOGIES RÉSEAUX

Cours : Système

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Niveau : L3

Date : 3 novembre 2026

Introduction

Dans le cadre du développement de compétences en administration du système d'exploitation Linux, la maîtrise de l'interface en ligne de commande et la compréhension de la gestion des environnements graphiques constituent des éléments majeurs. Ce rapport présente les travaux réalisés au cours du TD3, autour des thèmes suivants, Commandes de base, utilitaires, gestion des paquets et bureaux KDE Plasma et XFCE.

Ce TD a pour objectif principale d'acquérir une autonomie pratique dans la gestion quotidienne d'un système Linux. De la manipulation de bas niveau jusqu'à la configuration de l'interface utilisateur.

Étant divise en 5 grandes parties, dont la première vise à se repérer et manipuler des fichiers.

EXERCICE 1

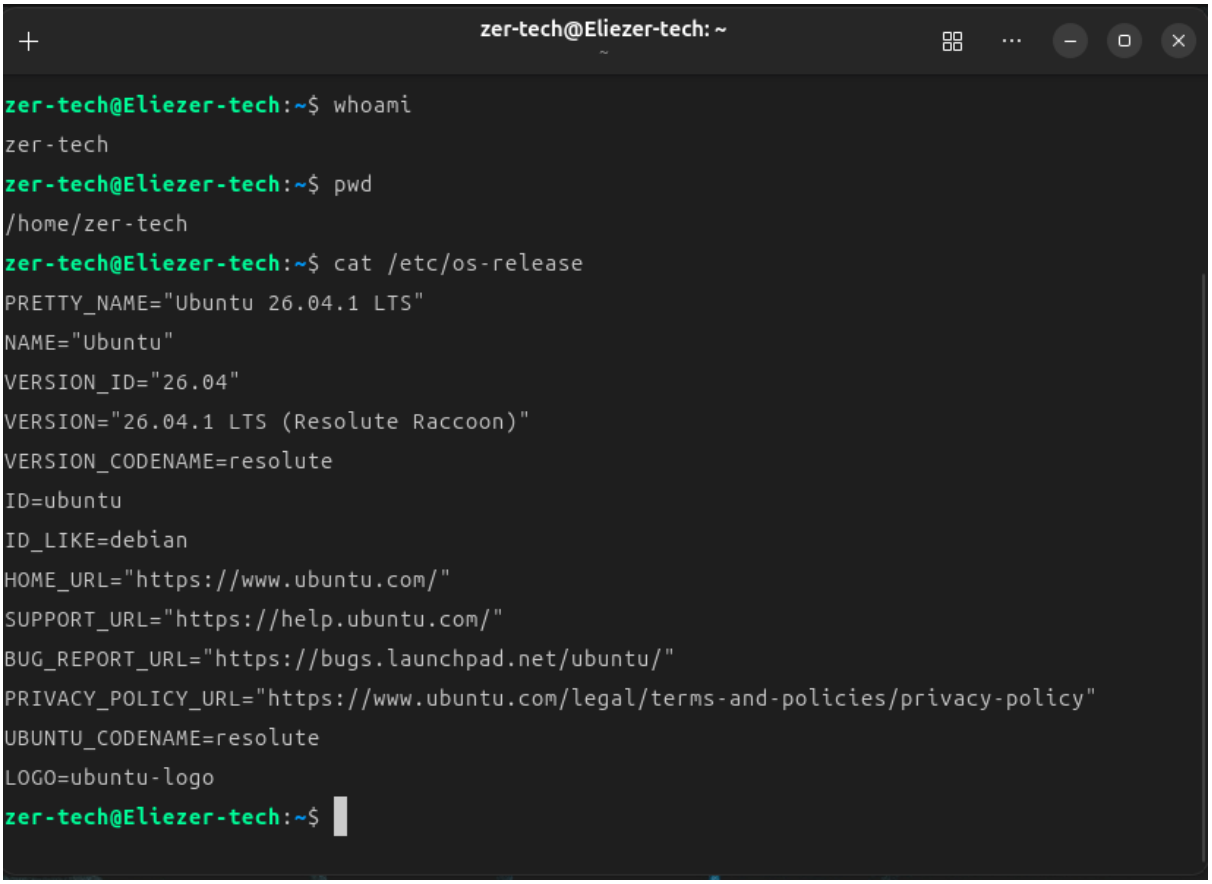
1- Afficher :

- Le nom de l'utilisateur
- Le chemin du répertoire courant
- Les informations sur la distribution
- La différence entre chemin absolu et chemin relatif.

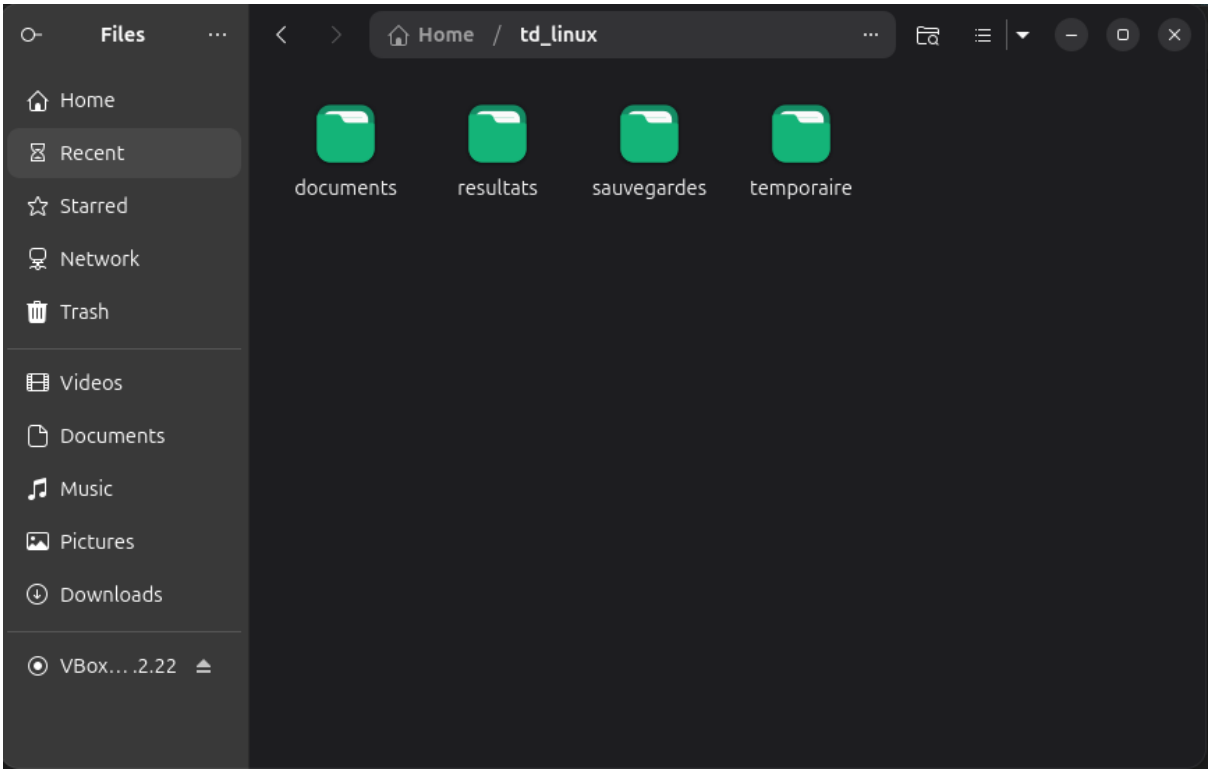
2- créer l'arborescence *~/td_linux(documents/sauvegardes/résultats)* et l'affiché au format détaillé

3-Créer un fichier *notes.txt*, le copier et le renommer dans *sauvegardes*

4- Créer et supprimer un fichier nommé *temporaire* puis faire la différence entre *rmdir et rm*



Les différents dossiers inclus le dossier temporaire sont bel et bien présent





Sous Linux, la différence réside dans le point de départ utilisé pour localiser un fichier ou un dossier.
Un chemin absolu donne l'adresse complète depuis la racine du système, tandis qu'un chemin relatif donne l'adresse par rapport à l'endroit où vous vous trouvez actuellement (le répertoire de travail).

rmdir (Remove Directory) : Elle sert uniquement à supprimer des répertoires vides. Si le dossier contient le moindre fichier ou sous-dossier, la commande échoue par sécurité pour éviter les pertes de données accidentelles.

rm (Remove) : Elle sert principalement à supprimer des fichiers. Cependant, en lui ajoutant des options (comme -r), elle devient capable de supprimer des dossiers et tout leur contenu (fichiers, sous-dossiers), qu'ils soient vides ou non.

EXERCICE 2

Dans la deuxième partie, vise à exploiter les utilitaires et faire du traitement de texte

La première tâche vise à créer un fichier nommé service.txt et :

1- Afficher

- Le contenu complet du ficher
- Ses deux premières et dernières lignes
- Et son nombre de lignes

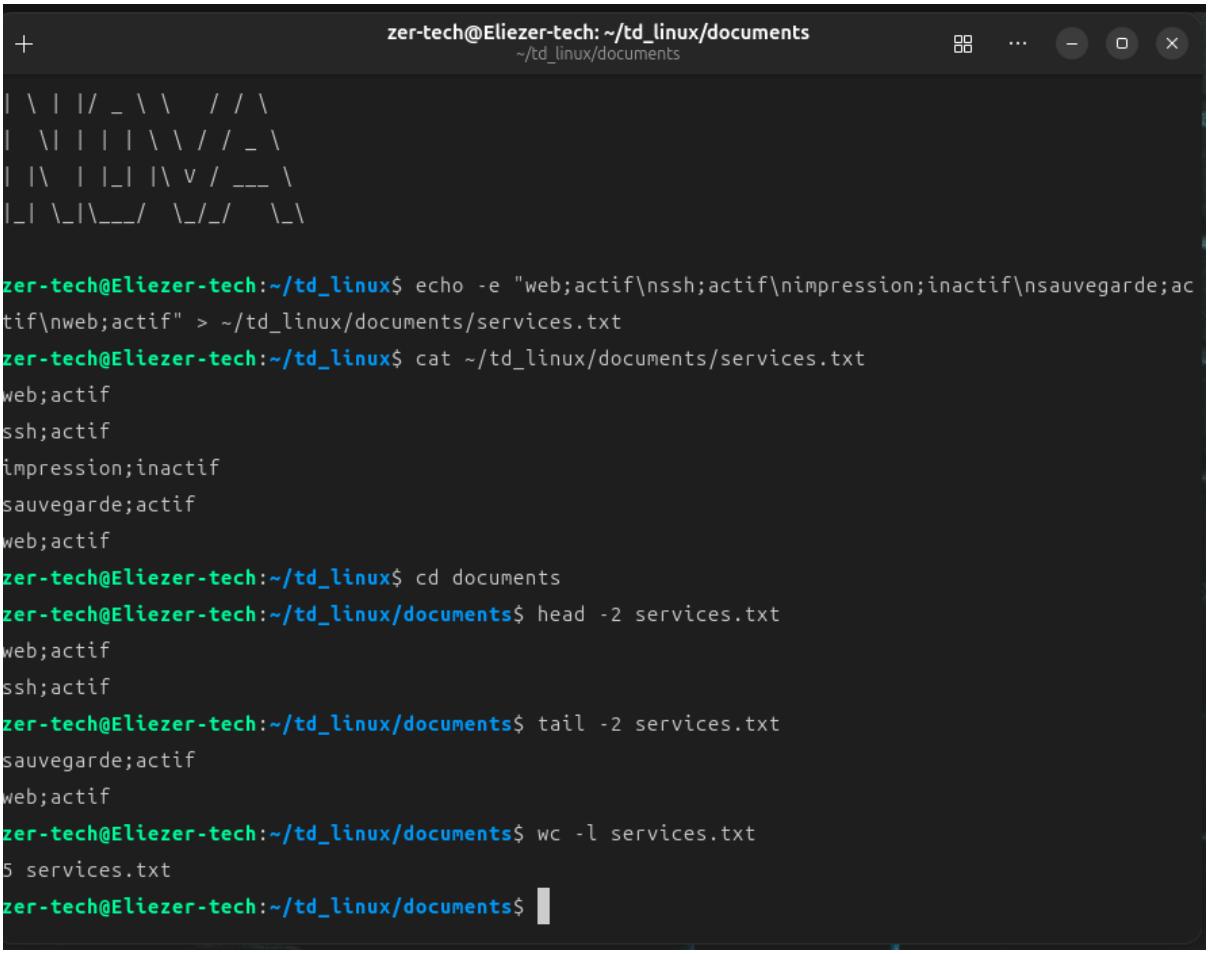
2- Extraire uniquement les lignes dont le deuxième champ vaut actif

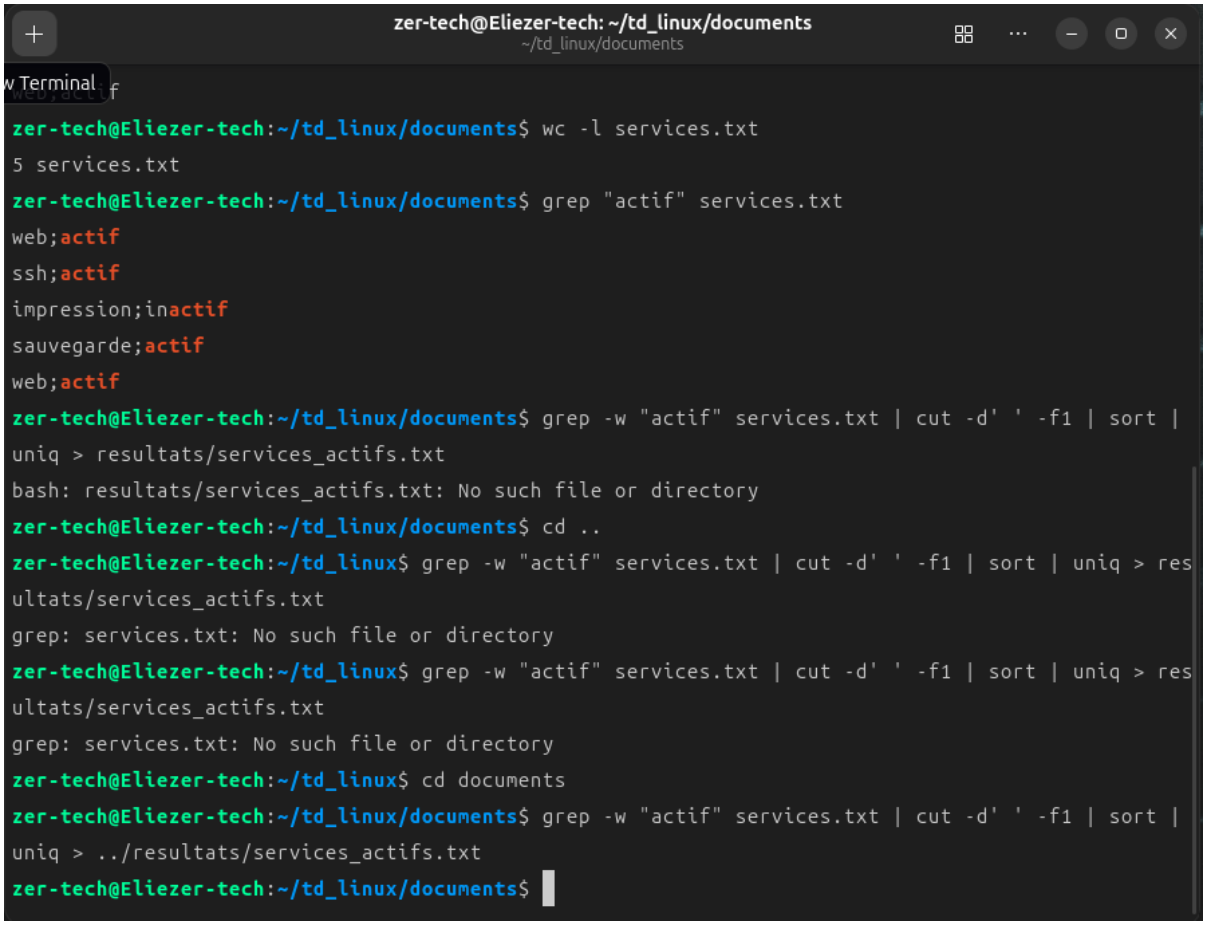
3- Avec une seule ligne de commande, sélectionner les services actifs, extraire leur nom, les trier et supprimer les doublons, et enregistrer le résultat dans résultats/services_actifs.txt

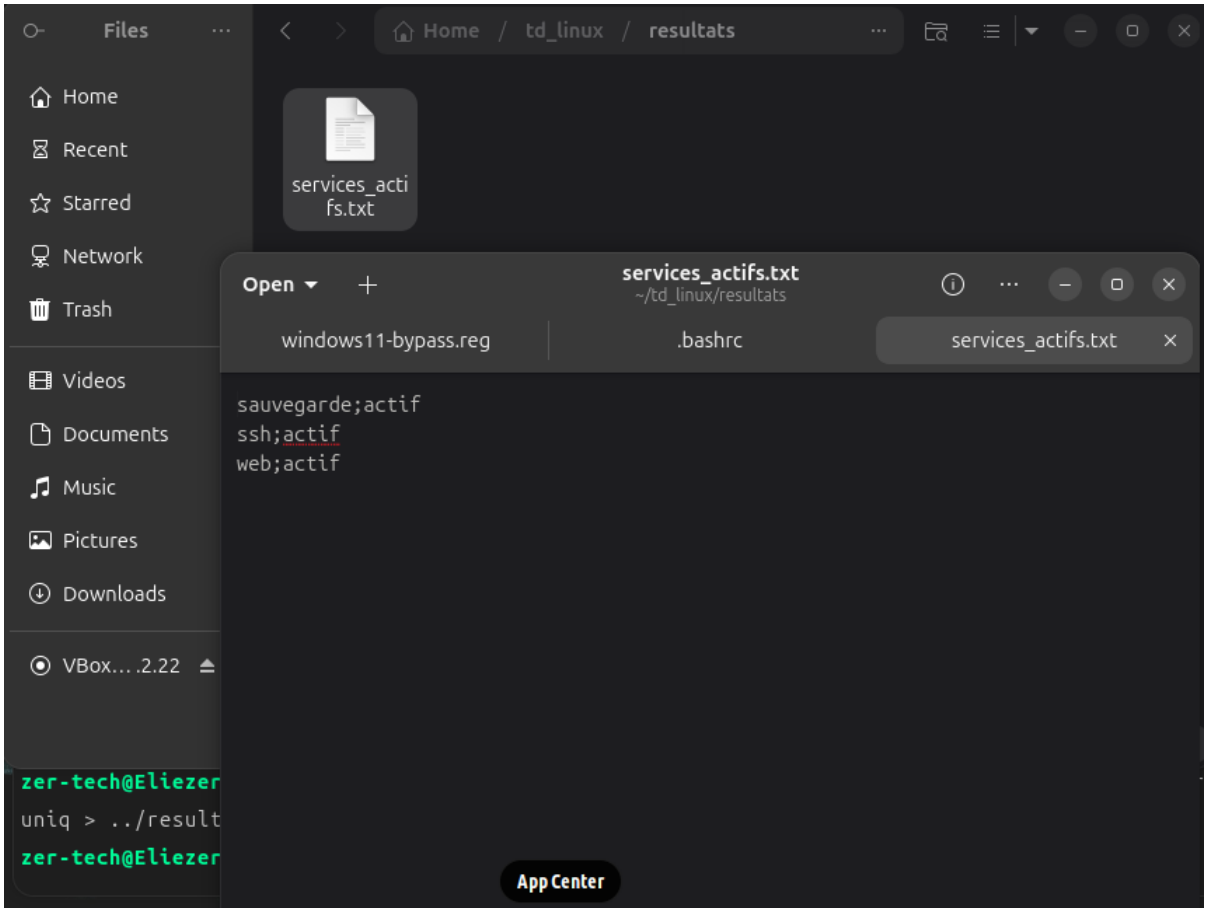
4- rechercher les fichiers avec l'extension .txt

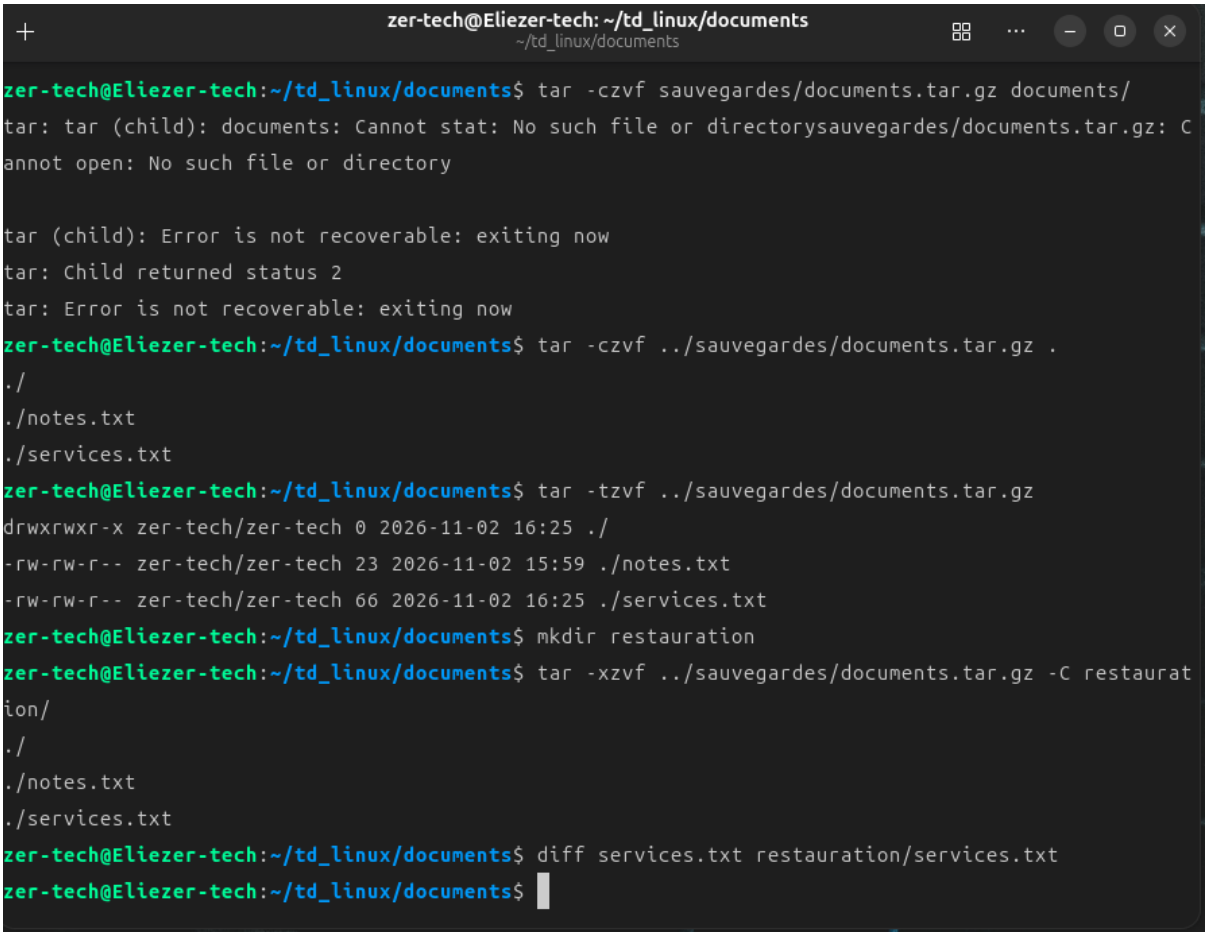
5- Créer l'archive sauvegardes/documents.tar.gz, afficher son contenu sans l'extraire, puis l'extraire dans un dossier restauration

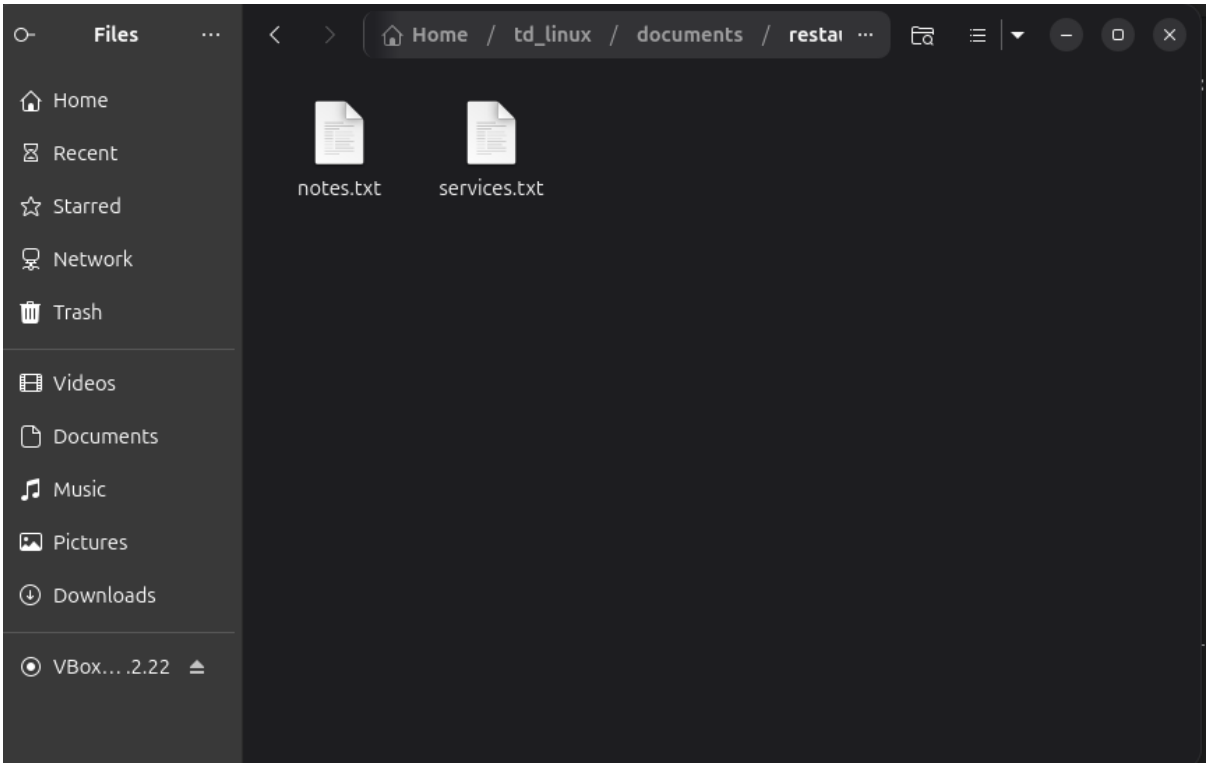
6- expliquer la différence entre les opérateurs > et >>




zer-tech@Eliezer-tech: ~/td_linux/documents
~/td_linux/documents
w Terminal
zer-tech@Eliezer-tech:~/td_linux/documents$ wc -l services.txt
5 services.txt
zer-tech@Eliezer-tech:~/td_linux/documents$ grep "actif" services.txt
web;actif
ssh;actif
impression;inactif
sauvegarde;actif
web;actif
zer-tech@Eliezer-tech:~/td_linux/documents$ grep -w "actif" services.txt | cut -d' ' -f1 | sort |
uniq > resultats/services_actifs.txt
bash: resultats/services_actifs.txt: No such file or directory
zer-tech@Eliezer-tech:~/td_linux/documents$ cd ..
zer-tech@Eliezer-tech:~/td_linux$ grep -w "actif" services.txt | cut -d' ' -f1 | sort | uniq > res
ultats/services_actifs.txt
grep: services.txt: No such file or directory
zer-tech@Eliezer-tech:~/td_linux$ grep -w "actif" services.txt | cut -d' ' -f1 | sort | uniq > res
ultats/services_actifs.txt
grep: services.txt: No such file or directory
zer-tech@Eliezer-tech:~/td_linux$ cd documents
zer-tech@Eliezer-tech:~/td_linux/documents$ grep -w "actif" services.txt | cut -d' ' -f1 | sort |
uniq > ../resultats/services_actifs.txt
zer-tech@Eliezer-tech:~/td_linux/documents$


Files
Home
Recent
Starred
Network
Trash
Videos
Documents
Music
Pictures
Downloads
VBox....2.22
Home / td_linux / resultats
services_actifs.txt
Open
services_actifs.txt
~/td_linux/resultats
windows11-bypass.reg
.bashrc
services_actifs.txt
sauvegarde;actif
ssh;actif
web;actif
zer-tech@Eliezer
uniq > ../result
zer-tech@Eliezer
App Center


zer-tech@Eliezer-tech: ~/td_linux/documents
~/td_linux/documents
zer-tech@Eliezer-tech:~/td_linux/documents$ tar -czvf sauvegardes/documents.tar.gz documents/
tar: tar (child): documents: Cannot stat: No such file or directorysauvegardes/documents.tar.gz: C
annot open: No such file or directory
tar (child): Error is not recoverable: exiting now
tar: Child returned status 2
tar: Error is not recoverable: exiting now
zer-tech@Eliezer-tech:~/td_linux/documents$ tar -czvf ../sauvegardes/documents.tar.gz .
./
./notes.txt
./services.txt
zer-tech@Eliezer-tech:~/td_linux/documents$ tar -tzvf ../sauvegardes/documents.tar.gz
drwxrwxr-x zer-tech/zer-tech 0 2026-11-02 16:25 ./
-rw-rw-r-- zer-tech/zer-tech 23 2026-11-02 15:59 ./notes.txt
-rw-rw-r-- zer-tech/zer-tech 66 2026-11-02 16:25 ./services.txt
zer-tech@Eliezer-tech:~/td_linux/documents$ mkdir restauration
zer-tech@Eliezer-tech:~/td_linux/documents$ tar -xzvf ../sauvegardes/documents.tar.gz -C restaurat
ion/
./
./notes.txt
./services.txt
zer-tech@Eliezer-tech:~/td_linux/documents$ diff services.txt restauration/services.txt
zer-tech@Eliezer-tech:~/td_linux/documents$



La différence entre l'opérateur > et >> réside dans le traitement du contenu existant lors de la redirection du résultat d'une commande vers un fichier texte.

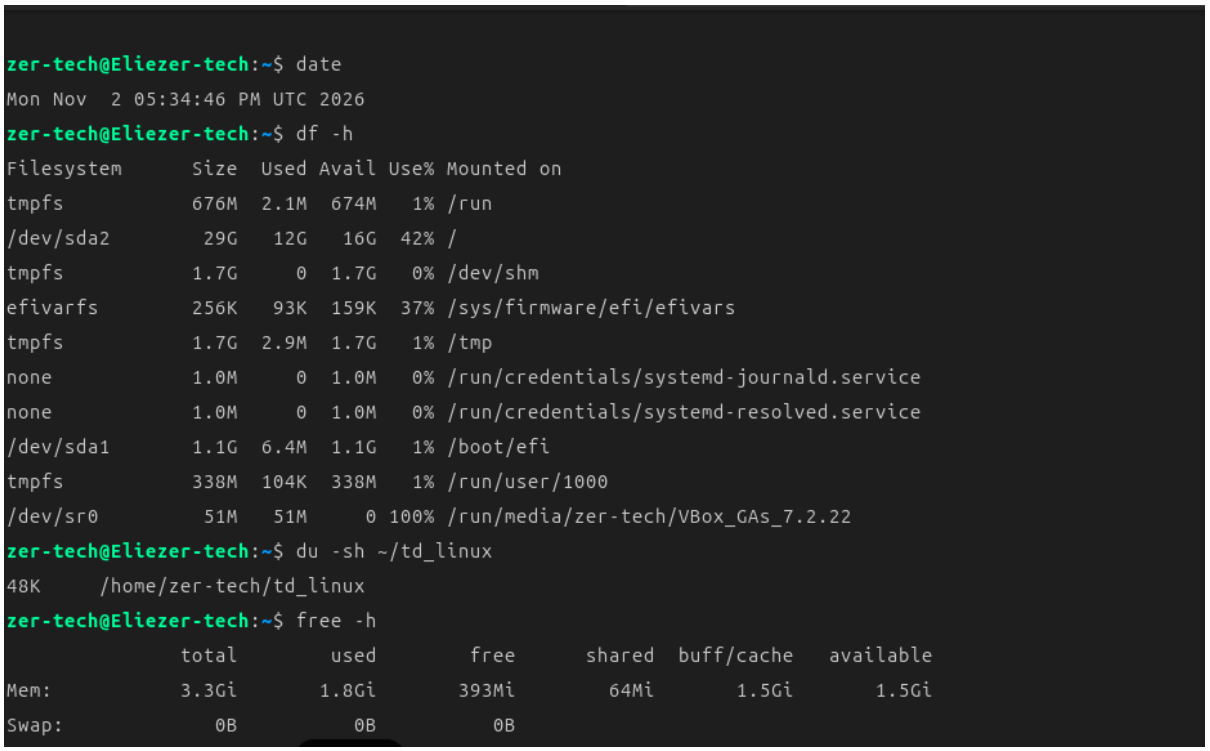
L'opérateur > écrase le contenu actuel du fichier, tandis que l'opérateur >> ajoute le contenu à la fin du fichier sans rien effacer.

EXERCICE 3

La troisième partie quant à elle vise à observer le système et les processus

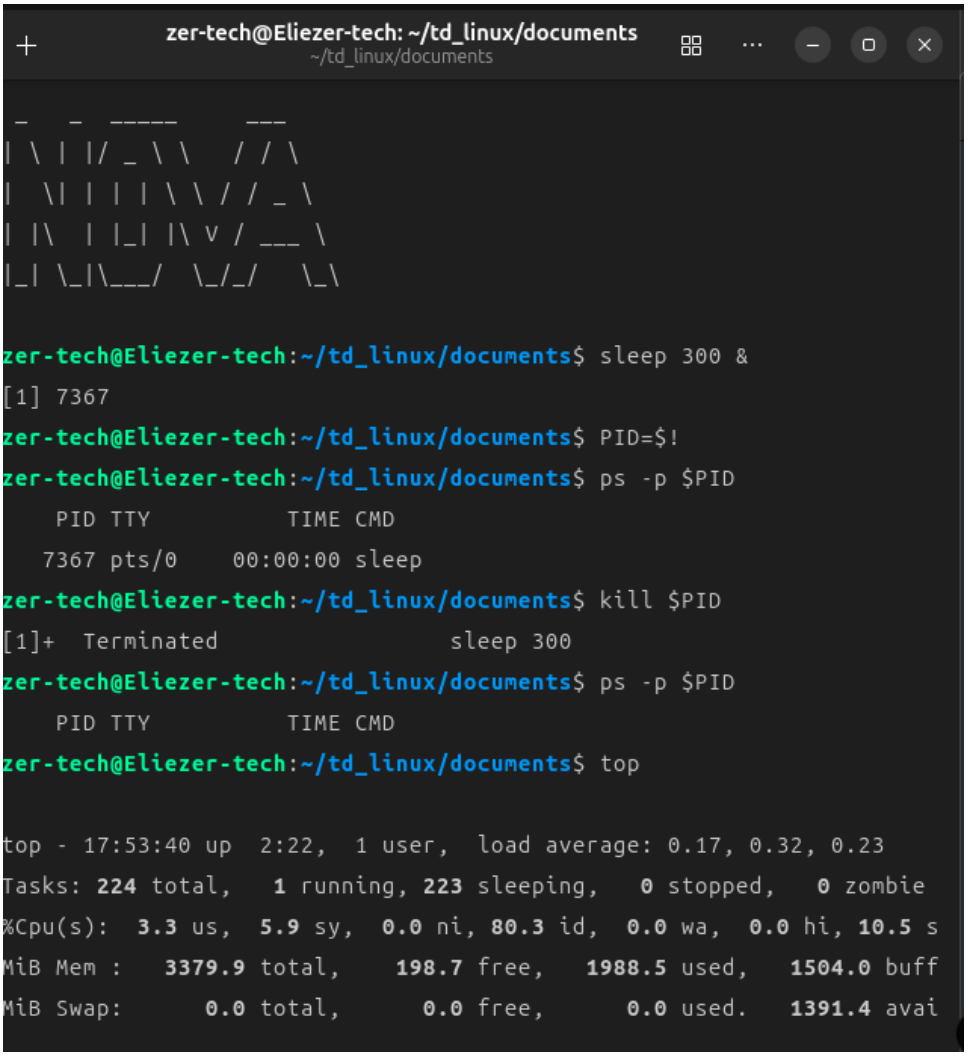
1. Afficher :
   - Expliquer la différence entre *df* et *du.*
   - La date et l'heure.
   - L'espace libre des systèmes de fichiers.
   - L'espace occupé par ~/td_linux.
   - La mémoire disponible.

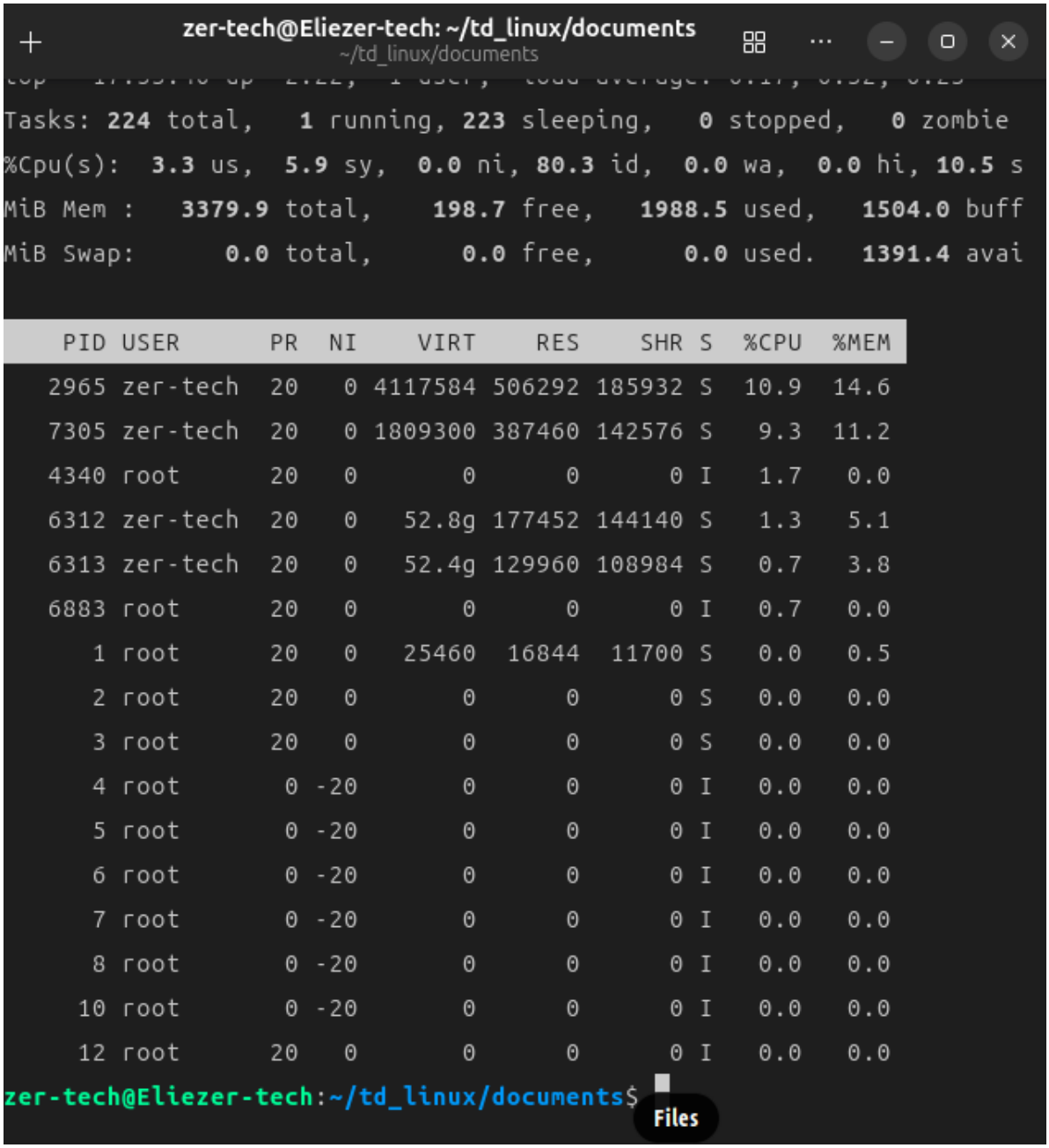
1. Lancer *sleep* 300 en arrière-plan. Relever immédiatement son PID avec *$!*, puis afficher ce processus avec *ps* .
2. Terminer uniquement ce processus avec *kill* et vérifier son arrêt.
3. Ouvrir *top*, relever un processus et expliquer ses consommations CPU et mémoire. Quitter avec *q.*



```
zer-tech@Eliezer-tech:~$ date
Mon Nov  2 05:34:46 PM UTC 2026
zer-tech@Eliezer-tech:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
tmpfs           676M  2.1M  674M   1% /run
/dev/sda2        29G   12G   16G  42% /
tmpfs           1.7G     0  1.7G   0% /dev/shm
efivarfs        256K   93K  159K  37% /sys/firmware/efi/efivars
tmpfs           1.7G  2.9M  1.7G   1% /tmp
none            1.0M     0  1.0M   0% /run/credentials/systemd-journald.service
none            1.0M     0  1.0M   0% /run/credentials/systemd-resolved.service
/dev/sda1       1.1G  6.4M  1.1G   1% /boot/efi
tmpfs           338M  104K  338M   1% /run/user/1000
/dev/sr0         51M   51M     0 100% /run/media/zer-tech/VBox_GAs_7.2.22
zer-tech@Eliezer-tech:~$ du -sh ~/td_linux
48K     /home/zer-tech/td_linux
zer-tech@Eliezer-tech:~$ free -h
               total        used        free      shared  buff/cache   available
Mem:           3.3Gi       1.8Gi       393Mi        64Mi       1.5Gi       1.5Gi
Swap:             0B          0B          0B
```

La commande *df* analyse l'ensemble du système de fichiers, tandis que la commande *de l'analyse* l'espace occupé par des fichiers ou dossiers précis (vision ciblée)

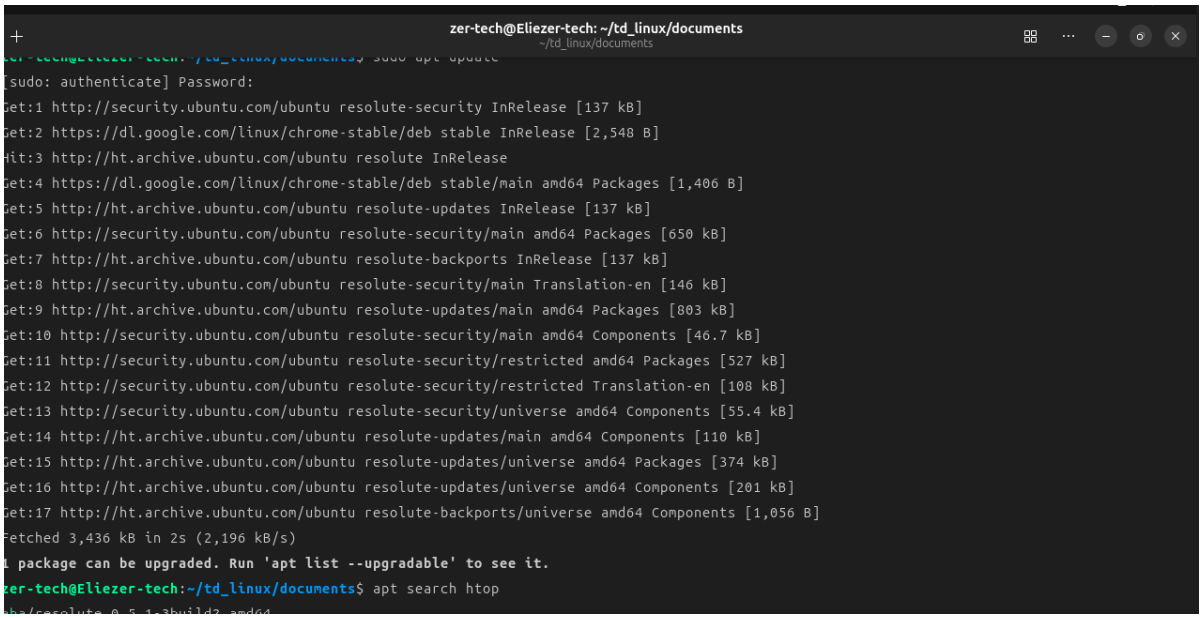

zer-tech@Eliezer-tech: ~/td_linux/documents
~/td_linux/documents
_ _ _____ ___
| \ | |/ _ \ \ / / \
| \| | | | \ \ / / _ \
| |\ | |_| |\ V / ___ \
|_| _|___/ _/_/ _\
zer-tech@Eliezer-tech:~/td_linux/documents$ sleep 300 &
[1] 7367
zer-tech@Eliezer-tech:~/td_linux/documents$ PID=$!
zer-tech@Eliezer-tech:~/td_linux/documents$ ps -p $PID
PID TTY TIME CMD
7367 pts/0 00:00:00 sleep
zer-tech@Eliezer-tech:~/td_linux/documents$ kill $PID
[1]+ Terminated sleep 300
zer-tech@Eliezer-tech:~/td_linux/documents$ ps -p $PID
PID TTY TIME CMD
zer-tech@Eliezer-tech:~/td_linux/documents$ top
top - 17:53:40 up 2:22, 1 user, load average: 0.17, 0.32, 0.23
Tasks: 224 total, 1 running, 223 sleeping, 0 stopped, 0 zombie
%Cpu(s): 3.3 us, 5.9 sy, 0.0 ni, 80.3 id, 0.0 wa, 0.0 hi, 10.5 s
MiB Mem : 3379.9 total, 198.7 free, 1988.5 used, 1504.0 buff
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 1391.4 avai

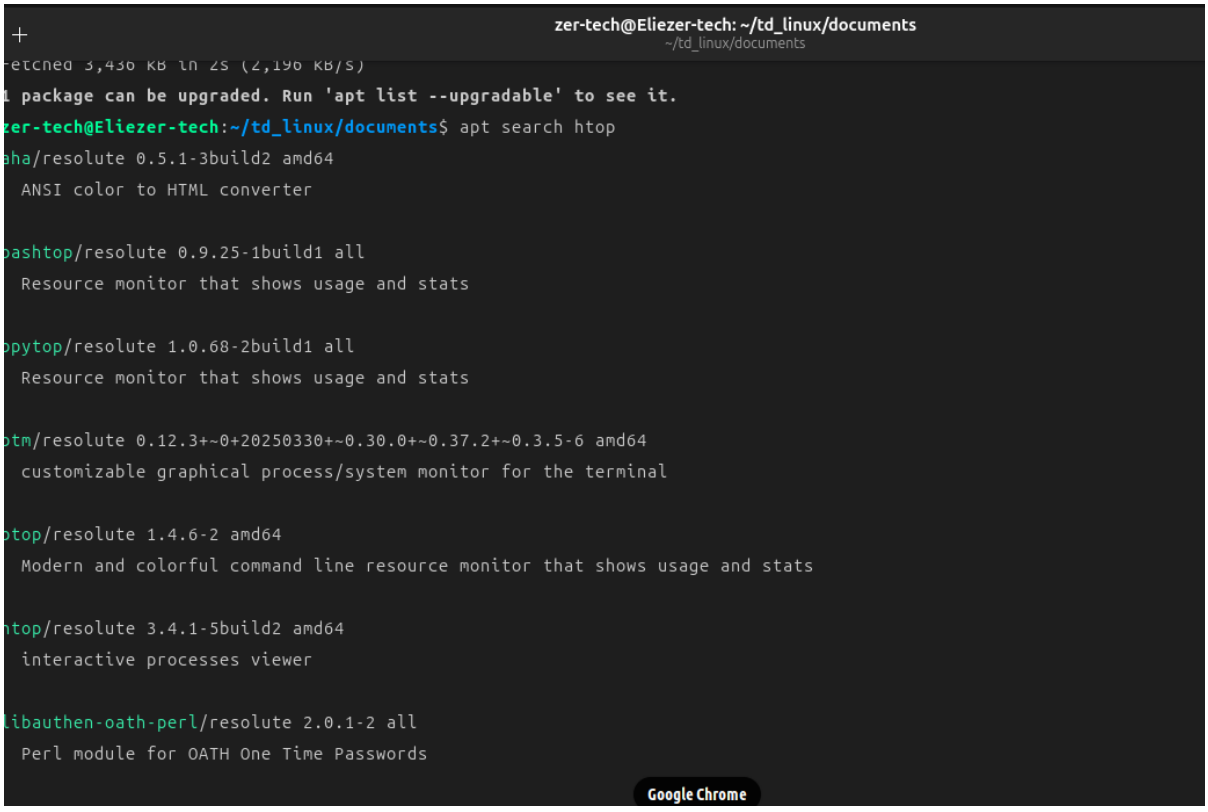

zer-tech@Eliezer-tech: ~/td_linux/documents
~/td_linux/documents
Tasks: 224 total, 1 running, 223 sleeping, 0 stopped, 0 zombie
%Cpu(s): 3.3 us, 5.9 sy, 0.0 ni, 80.3 id, 0.0 wa, 0.0 hi, 10.5 s
MiB Mem : 3379.9 total, 198.7 free, 1988.5 used, 1504.0 buff
MiB Swap: 0.0 total, 0.0 free, 0.0 used. 1391.4 avai
PID USER PR NI VIRT RES SHR S %CPU %MEM
2965 zer-tech 20 0 4117584 506292 185932 S 10.9 14.6
7305 zer-tech 20 0 1809300 387460 142576 S 9.3 11.2
4340 root 20 0 0 0 0 I 1.7 0.0
6312 zer-tech 20 0 52.8g 177452 144140 S 1.3 5.1
6313 zer-tech 20 0 52.4g 129960 108984 S 0.7 3.8
6883 root 20 0 0 0 0 I 0.7 0.0
1 root 20 0 25460 16844 11700 S 0.0 0.5
2 root 20 0 0 0 0 S 0.0 0.0
3 root 20 0 0 0 0 S 0.0 0.0
4 root 0 -20 0 0 0 I 0.0 0.0
5 root 0 -20 0 0 0 I 0.0 0.0
6 root 0 -20 0 0 0 I 0.0 0.0
7 root 0 -20 0 0 0 I 0.0 0.0
8 root 0 -20 0 0 0 I 0.0 0.0
10 root 0 -20 0 0 0 I 0.0 0.0
12 root 20 0 0 0 0 I 0.0 0.0
zer-tech@Eliezer-tech:~/td_linux/documents$
Files

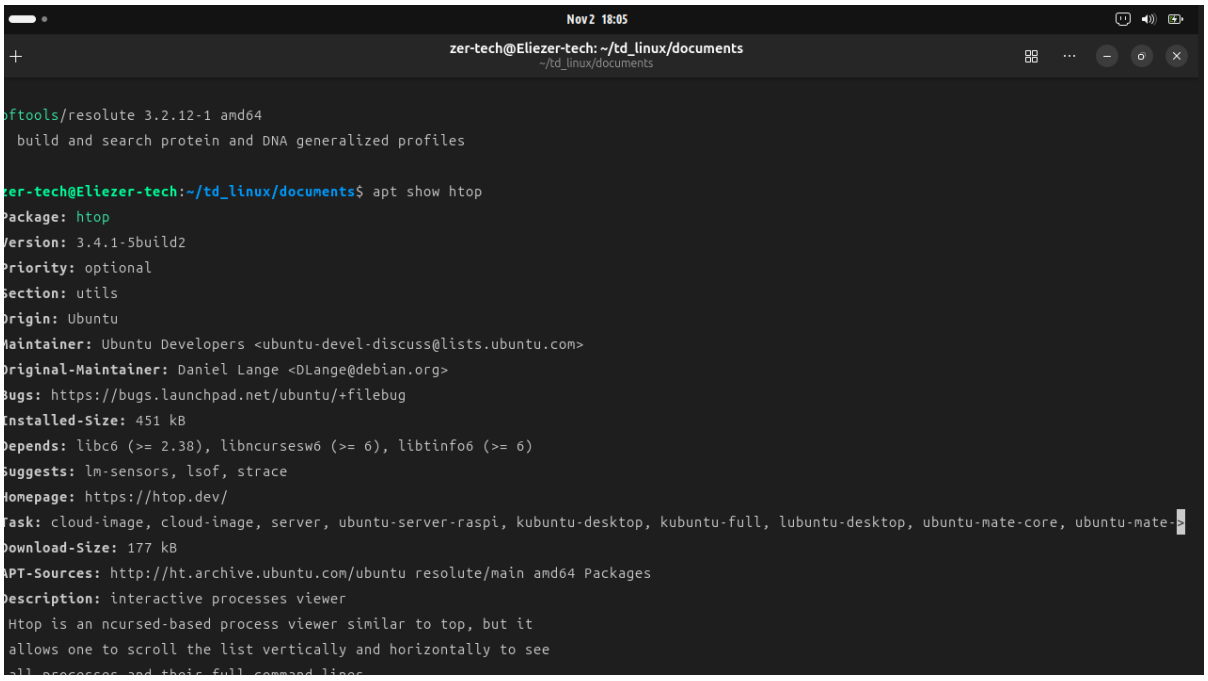
EXERCICE 4

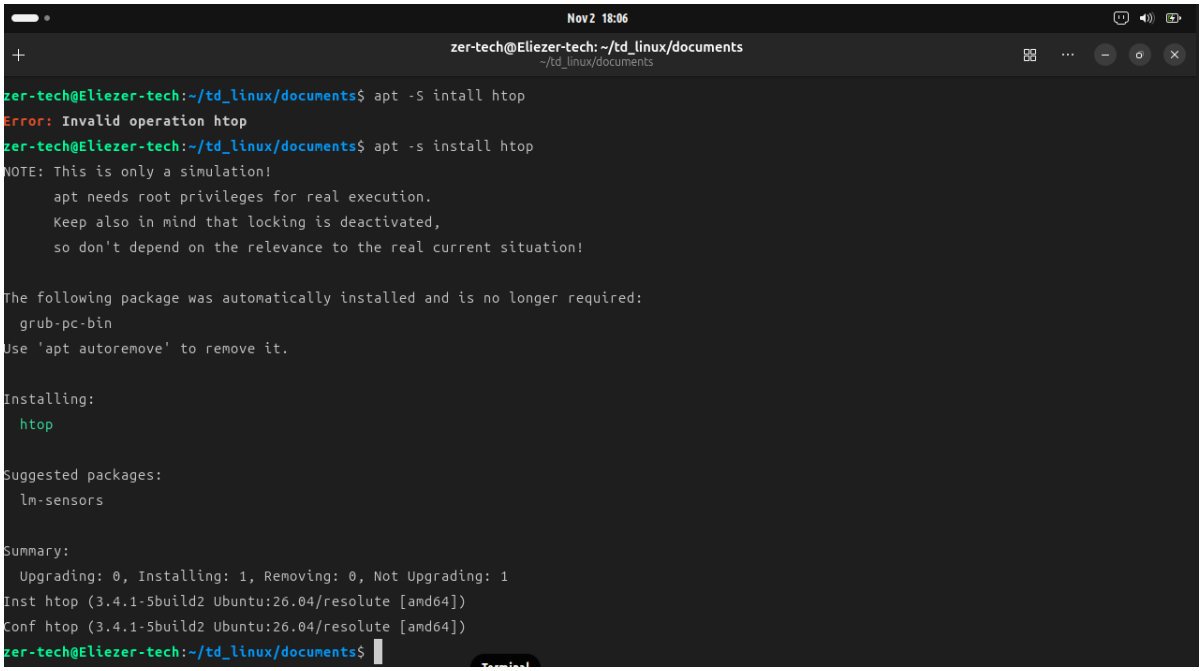
La partie 4 vise à faire l'installation et la gestion des packets

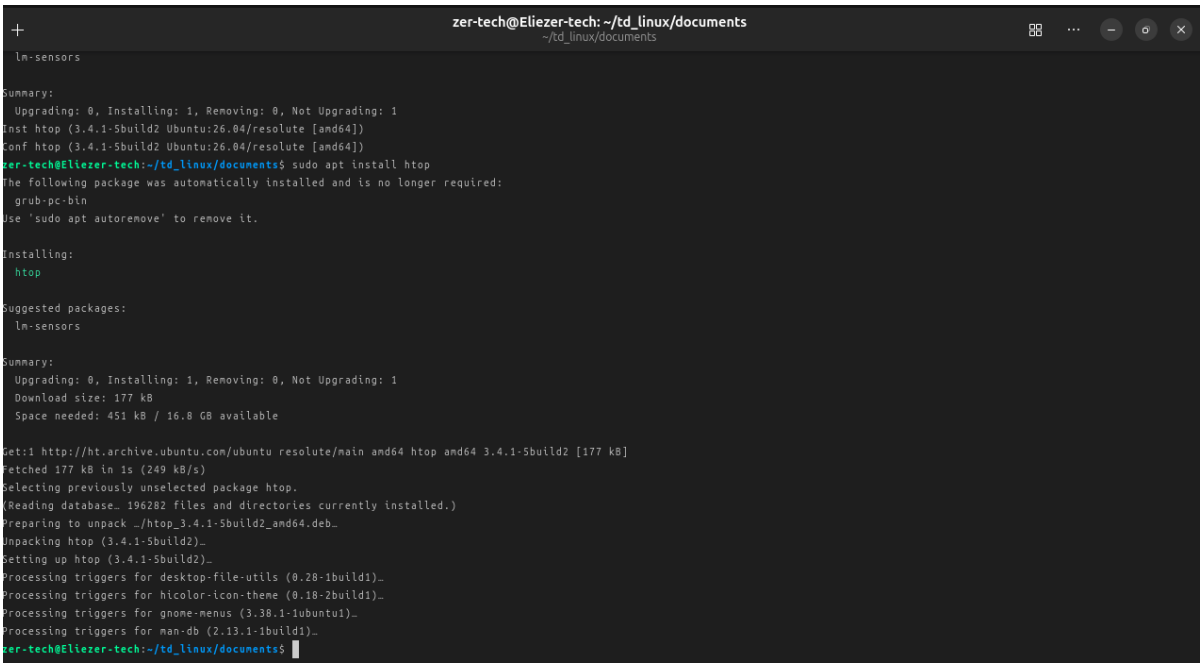
Sous Linux la différence réside dans l'action effectuée. En effet, tandis que l'une met à jour la liste des logiciels disponibles, l'autre installe concrètement les nouvelles versions. Donc *sudo apt update actualise l'index* des paquets, tandis que *sudo apt upgrade* télécharge et installe les mises à jour logicielles sur la machine.

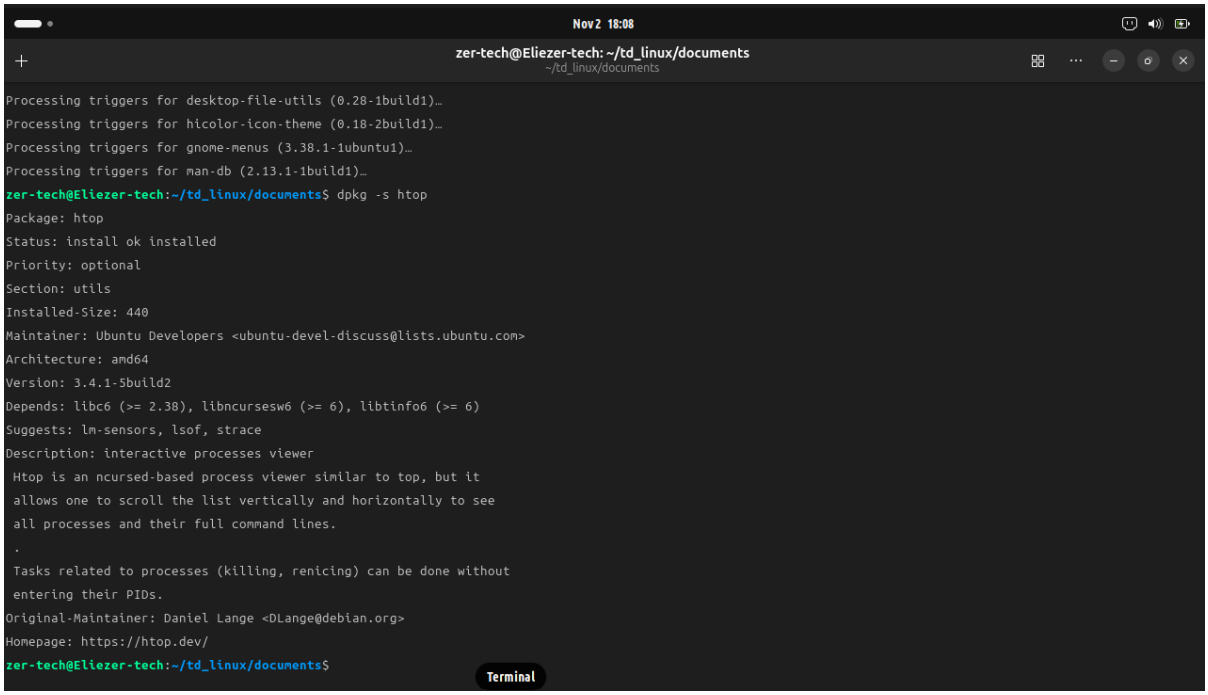


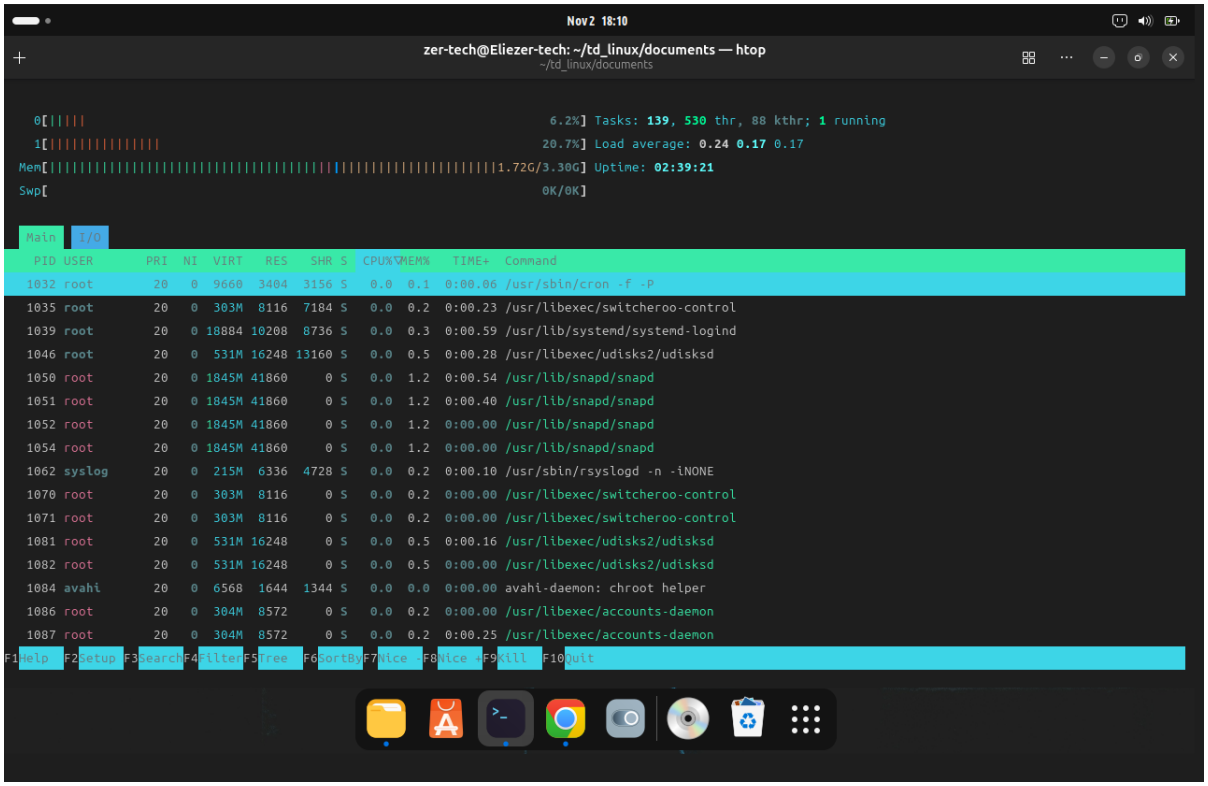

zer-tech@Eliezer-tech: ~/td_linux/documents
~/td_linux/documents
Fetched 3,436 kB in 2s (2,196 kB/s)
1 package can be upgraded. Run 'apt list --upgradable' to see it.
zer-tech@Eliezer-tech:~/td_linux/documents$ apt search htop
aha/resolute 0.5.1-3build2 amd64
ANSI color to HTML converter
bashtop/resolute 0.9.25-1build1 all
Resource monitor that shows usage and stats
bpytop/resolute 1.0.68-2build1 all
Resource monitor that shows usage and stats
btm/resolute 0.12.3+~0+20250330+~0.30.0+~0.37.2+~0.3.5-6 amd64
customizable graphical process/system monitor for the terminal
btop/resolute 1.4.6-2 amd64
Modern and colorful command line resource monitor that shows usage and stats
htop/resolute 3.4.1-5build2 amd64
interactive processes viewer
libauthen-oath-perl/resolute 2.0.1-2 all
Perl module for OATH One Time Passwords
Google Chrome

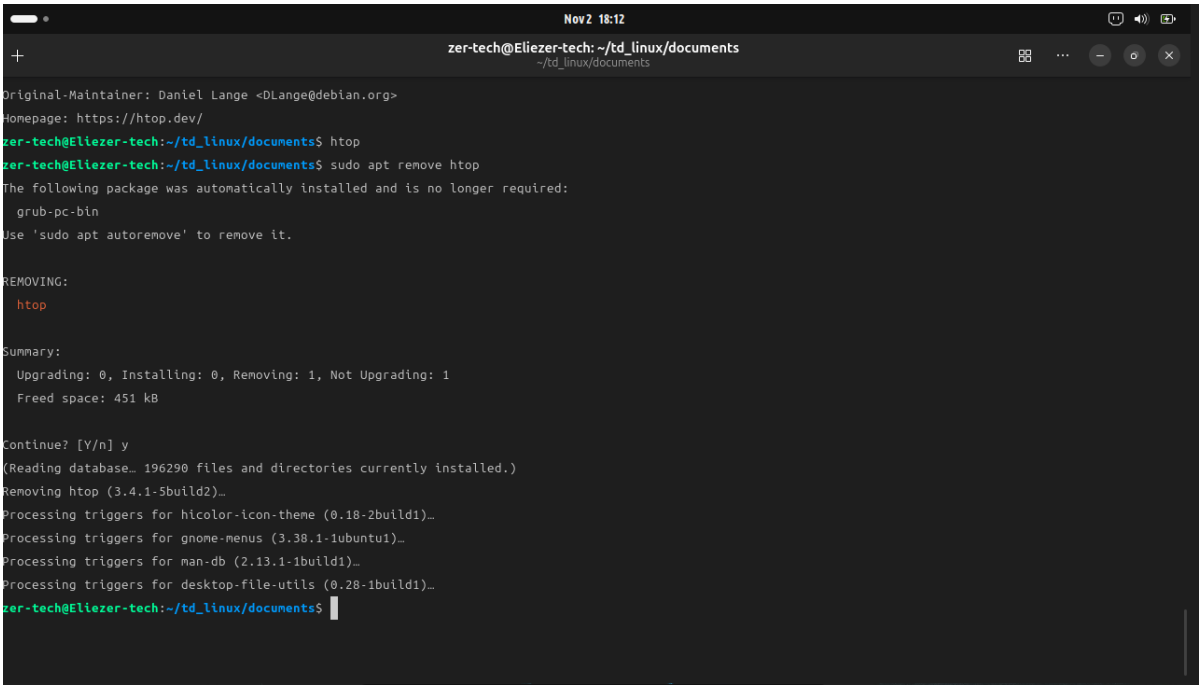

Nov 2 18:05
zer-tech@Eliezer-tech: ~/td_linux/documents
~/td_linux/documents
bftools/resolute 3.2.12-1 amd64
build and search protein and DNA generalized profiles
zer-tech@Eliezer-tech:~/td_linux/documents$ apt show htop
Package: htop
Version: 3.4.1-5build2
Priority: optional
Section: utils
Origin: Ubuntu
Maintainer: Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>
Original-Maintainer: Daniel Lange <DLange@debian.org>
Bugs: https://bugs.launchpad.net/ubuntu/+filebug
Installed-Size: 451 kB
Depends: libc6 (>= 2.38), libncursesw6 (>= 6), libtinfo6 (>= 6)
Suggests: lm-sensors, lsof, strace
Homepage: https://htop.dev/
Task: cloud-image, cloud-image, server, ubuntu-server-raspi, kubuntu-desktop, kubuntu-full, lubuntu-desktop, ubuntu-mate-core, ubuntu-mate-
Download-Size: 177 kB
APT-Sources: http://ht.archive.ubuntu.com/ubuntu resolute/main amd64 Packages
Description: interactive processes viewer
Htop is an ncursed-based process viewer similar to top, but it
allows one to scroll the list vertically and horizontally to see

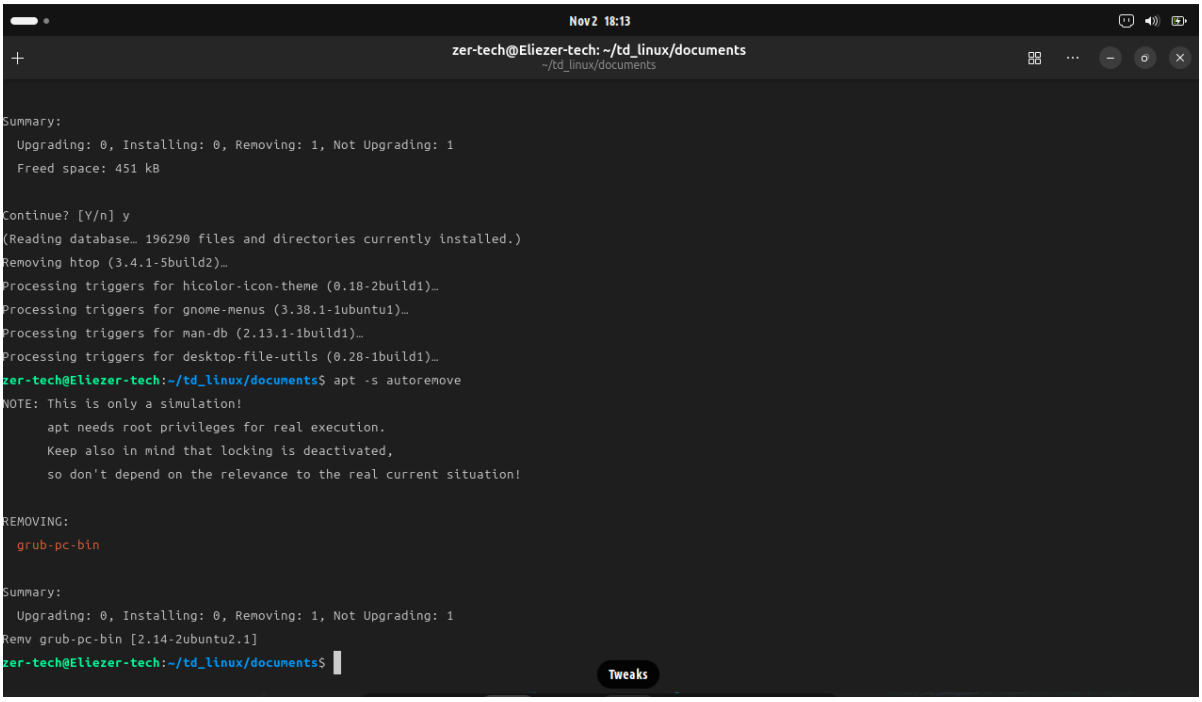

Nov 2 18:06
zer-tech@Eliezer-tech: ~/td_linux/documents
~/td_linux/documents
zer-tech@Eliezer-tech:~/td_linux/documents$ apt -S intall htop
Error: Invalid operation htop
zer-tech@Eliezer-tech:~/td_linux/documents$ apt -s install htop
NOTE: This is only a simulation!
apt needs root privileges for real execution.
Keep also in mind that locking is deactivated,
so don't depend on the relevance to the real current situation!
The following package was automatically installed and is no longer required:
grub-pc-bin
Use 'apt autoremove' to remove it.
Installing:
htop
Suggested packages:
lm-sensors
Summary:
Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1
Inst htop (3.4.1-5build2 Ubuntu:26.04/resolute [amd64])
Conf htop (3.4.1-5build2 Ubuntu:26.04/resolute [amd64])
zer-tech@Eliezer-tech:~/td_linux/documents$


zer-tech@Eliezer-tech: ~/td_linux/documents
~/td_linux/documents
lm-sensors
Summary:
Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1
Inst htop (3.4.1-5build2 Ubuntu:26.04/resolute [amd64])
Conf htop (3.4.1-5build2 Ubuntu:26.04/resolute [amd64])
zer-tech@Eliezer-tech:~/td_linux/documents$ sudo apt install htop
The following package was automatically installed and is no longer required:
grub-pc-bin
Use 'sudo apt autoremove' to remove it.
Installing:
htop
Suggested packages:
lm-sensors
Summary:
Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1
Download size: 177 kB
Space needed: 451 kB / 16.8 GB available
Get:1 http://ht.archive.ubuntu.com/ubuntu resolute/main amd64 htop amd64 3.4.1-5build2 [177 kB]
Fetched 177 kB in 1s (249 kB/s)
Selecting previously unselected package htop.
(Reading database… 196282 files and directories currently installed.)
Preparing to unpack …/htop_3.4.1-5build2_amd64.deb…
Unpacking htop (3.4.1-5build2)…
Setting up htop (3.4.1-5build2)…
Processing triggers for desktop-file-utils (0.28-1build1)…
Processing triggers for hicolor-icon-theme (0.18-2build1)…
Processing triggers for gnome-menus (3.38.1-1ubuntu1)…
Processing triggers for man-db (2.13.1-1build1)…
zer-tech@Eliezer-tech:~/td_linux/documents$


Nov 2 18:08
zer-tech@Eliezer-tech: ~/td_linux/documents
~/td_linux/documents
Processing triggers for desktop-file-utils (0.28-1build1)…
Processing triggers for hicolor-icon-theme (0.18-2build1)…
Processing triggers for gnome-menus (3.38.1-1ubuntu1)…
Processing triggers for man-db (2.13.1-1build1)…
zer-tech@Eliezer-tech:~/td_linux/documents$ dpkg -s htop
Package: htop
Status: install ok installed
Priority: optional
Section: utils
Installed-Size: 440
Maintainer: Ubuntu Developers <ubuntu-devel-discuss@lists.ubuntu.com>
Architecture: amd64
Version: 3.4.1-5build2
Depends: libc6 (>= 2.38), libncursesw6 (>= 6), libtinfo6 (>= 6)
Suggests: lm-sensors, lsof, strace
Description: interactive processes viewer
Htop is an ncursed-based process viewer similar to top, but it
allows one to scroll the list vertically and horizontally to see
all processes and their full command lines.
.
Tasks related to processes (killing, renicing) can be done without
entering their PIDs.
Original-Maintainer: Daniel Lange <DLange@debian.org>
Homepage: https://htop.dev/
zer-tech@Eliezer-tech:~/td_linux/documents$
Terminal


Nov 2 18:10
zer-tech@Eliezer-tech: ~/td_linux/documents — htop
~/td_linux/documents
0[||||| 6.2%] Tasks: 139, 530 thr, 88 kthr; 1 running
1[||||||||||||||| 20.7%] Load average: 0.24 0.17
Mem[|||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||||1.72G/3.30G] Uptime: 02:39:21
Swp[0K/0K]
Main I/O
PID USER PRI NI VIRT RES SHR S CPU% MEM% TIME+ Command
1032 root 20 0 9660 3404 3156 S 0.0 0.1 0:00.06 /usr/sbin/cron -f -P
1035 root 20 0 303M 8116 7184 S 0.0 0.2 0:00.23 /usr/libexec/switcheroo-control
1039 root 20 0 18884 10208 8736 S 0.0 0.3 0:00.59 /usr/lib/systemd/systemd-logind
1046 root 20 0 531M 16248 13160 S 0.0 0.5 0:00.28 /usr/libexec/udisks2/udisksd
1050 root 20 0 1845M 41860 0 S 0.0 1.2 0:00.54 /usr/lib/snapd/snapd
1051 root 20 0 1845M 41860 0 S 0.0 1.2 0:00.40 /usr/lib/snapd/snapd
1052 root 20 0 1845M 41860 0 S 0.0 1.2 0:00.00 /usr/lib/snapd/snapd
1054 root 20 0 1845M 41860 0 S 0.0 1.2 0:00.00 /usr/lib/snapd/snapd
1062 syslog 20 0 215M 6336 4728 S 0.0 0.2 0:00.10 /usr/sbin/rsyslogd -n -iNONE
1070 root 20 0 303M 8116 0 S 0.0 0.2 0:00.00 /usr/libexec/switcheroo-control
1071 root 20 0 303M 8116 0 S 0.0 0.2 0:00.00 /usr/libexec/switcheroo-control
1081 root 20 0 531M 16248 0 S 0.0 0.5 0:00.16 /usr/libexec/udisks2/udisksd
1082 root 20 0 531M 16248 0 S 0.0 0.5 0:00.00 /usr/libexec/udisks2/udisksd
1084 avahi 20 0 6568 1644 1344 S 0.0 0.0 0:00.00 avahi-daemon: chroot helper
1086 root 20 0 304M 8572 0 S 0.0 0.2 0:00.00 /usr/libexec/accounts-daemon
1087 root 20 0 304M 8572 0 S 0.0 0.2 0:00.25 /usr/libexec/accounts-daemon
F1Help F2Setup F3Search F4Filter F5Tree F6SortBy F7Nice - F8Nice + F9Kill F10Quit


Nov 2 18:12
zer-tech@Eliezer-tech: ~/td_linux/documents
~/td_linux/documents
Original-Maintainer: Daniel Lange <DLange@debian.org>
Homepage: https://htop.dev/
zer-tech@Eliezer-tech:~/td_linux/documents$ htop
zer-tech@Eliezer-tech:~/td_linux/documents$ sudo apt remove htop
The following package was automatically installed and is no longer required:
grub-pc-bin
Use 'sudo apt autoremove' to remove it.
REMOVING:
htop
Summary:
Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 1
Freed space: 451 kB
Continue? [Y/n] y
(Reading database… 196290 files and directories currently installed.)
Removing htop (3.4.1-5build2)…
Processing triggers for hicolor-icon-theme (0.18-2build1)…
Processing triggers for gnome-menus (3.38.1-1ubuntu1)…
Processing triggers for man-db (2.13.1-1build1)…
Processing triggers for desktop-file-utils (0.28-1build1)…
zer-tech@Eliezer-tech:~/td_linux/documents$


Nov 2 18:13
zer-tech@Eliezer-tech: ~/td_linux/documents
~/td_linux/documents
Summary:
Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 1
Freed space: 451 kB
Continue? [Y/n] y
(Reading database… 196290 files and directories currently installed.)
Removing htop (3.4.1-5build2)…
Processing triggers for hicolor-icon-theme (0.18-2build1)…
Processing triggers for gnome-menus (3.38.1-1ubuntu1)…
Processing triggers for man-db (2.13.1-1build1)…
Processing triggers for desktop-file-utils (0.28-1build1)…
zer-tech@Eliezer-tech:~/td_linux/documents$ apt -s autoremove
NOTE: This is only a simulation!
apt needs root privileges for real execution.
Keep also in mind that locking is deactivated,
so don't depend on the relevance to the real current situation!
REMOVING:
grub-pc-bin
Summary:
Upgrading: 0, Installing: 0, Removing: 1, Not Upgrading: 1
Remv grub-pc-bin [2.14-2ubuntu2.1]
zer-tech@Eliezer-tech:~/td_linux/documents$
Tweaks

Exécutons ces commandes

ping google.com # Tester la connectivité réseau

ip a # Affiche les interfaces réseau et IP

history # Affiche l'historique des commandes

cowsay "Bienvenue dans le cours Linux" # Le terminal te parle avec une vache

cmatrix # Effet "Matrix" dans le terminal

neofetch # Affiche les infos système avec logo Linux

journalctl -xe # Consulter les logs système récents

sudo dmesg

sudo dmesg | grep -i memory

date

cal

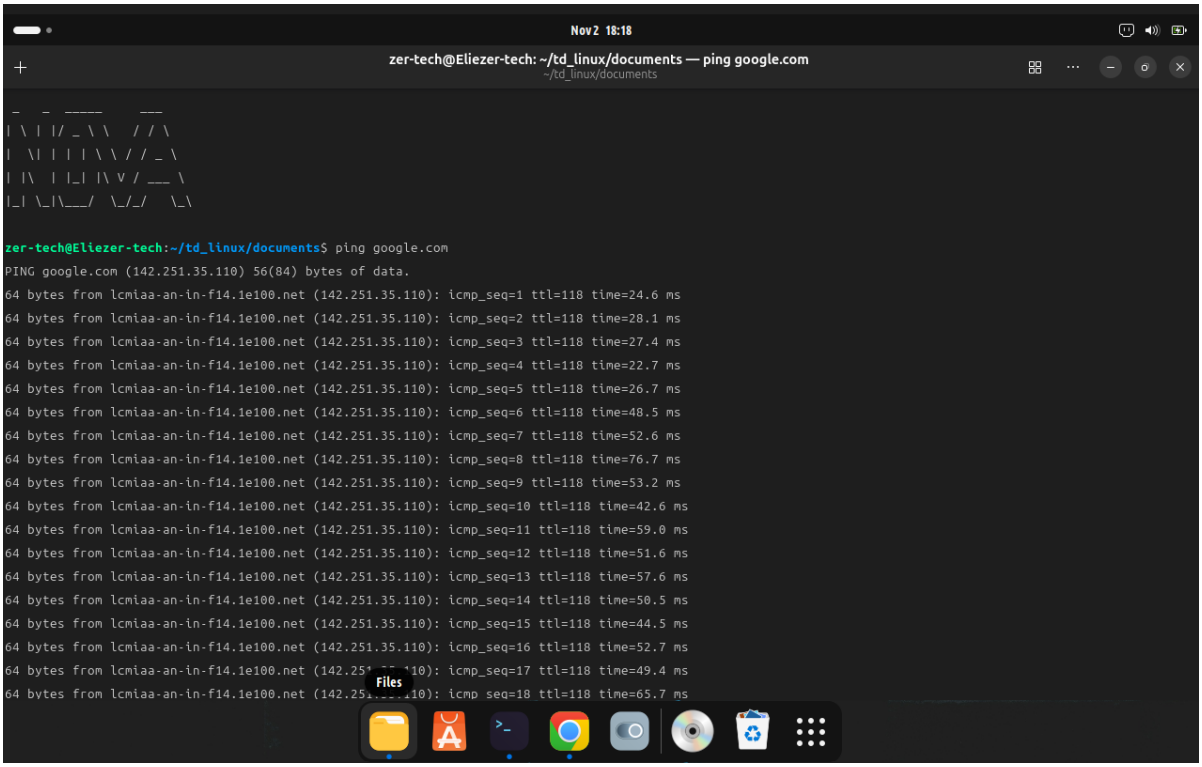
figlet "UBUNTU"# Écrire un grand titre

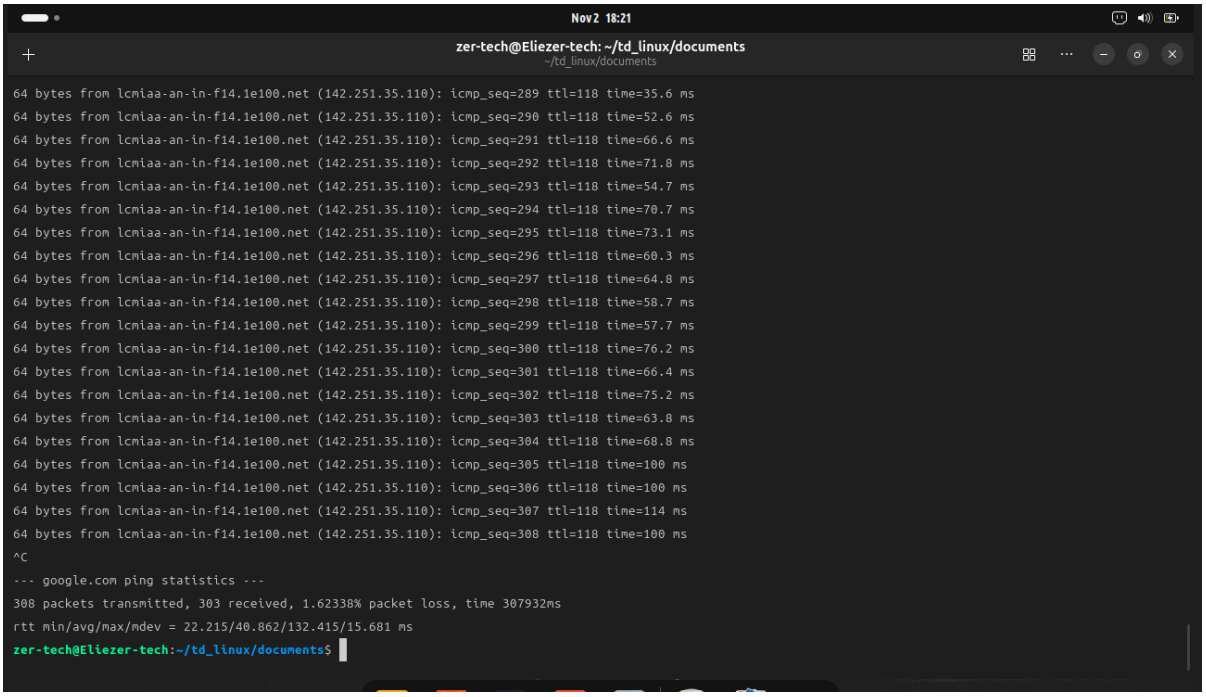
sl # Écrire un grand titre

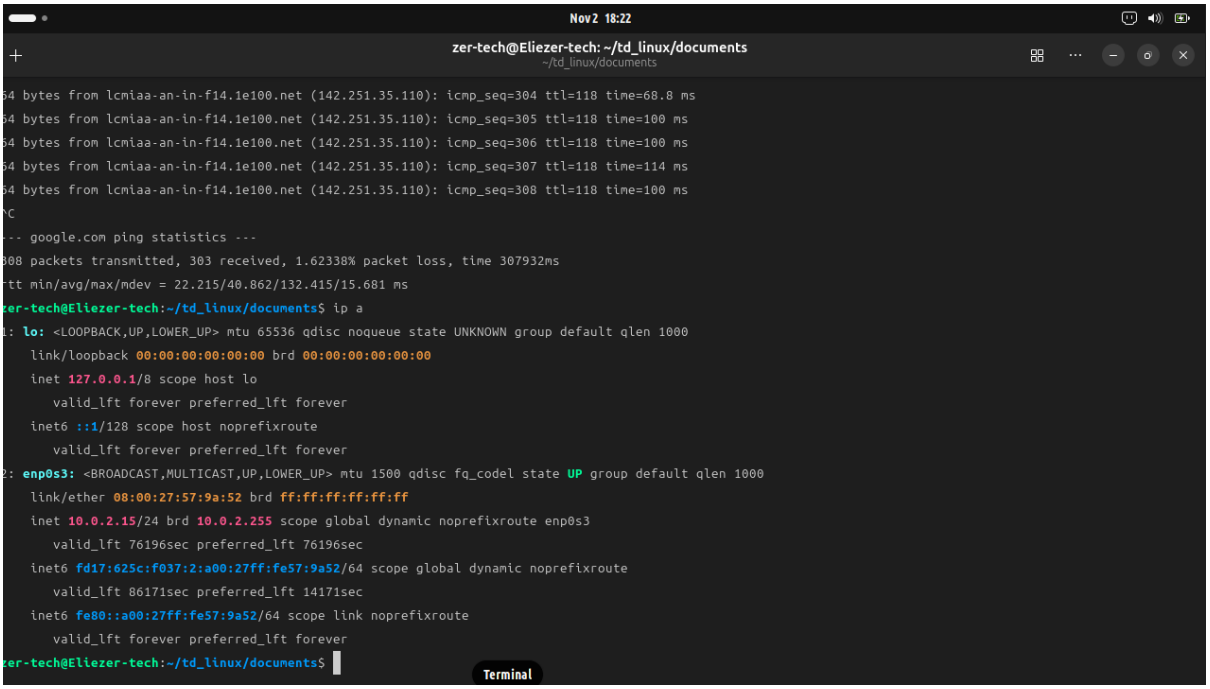
seq 1 10 # Écrire un grand titre

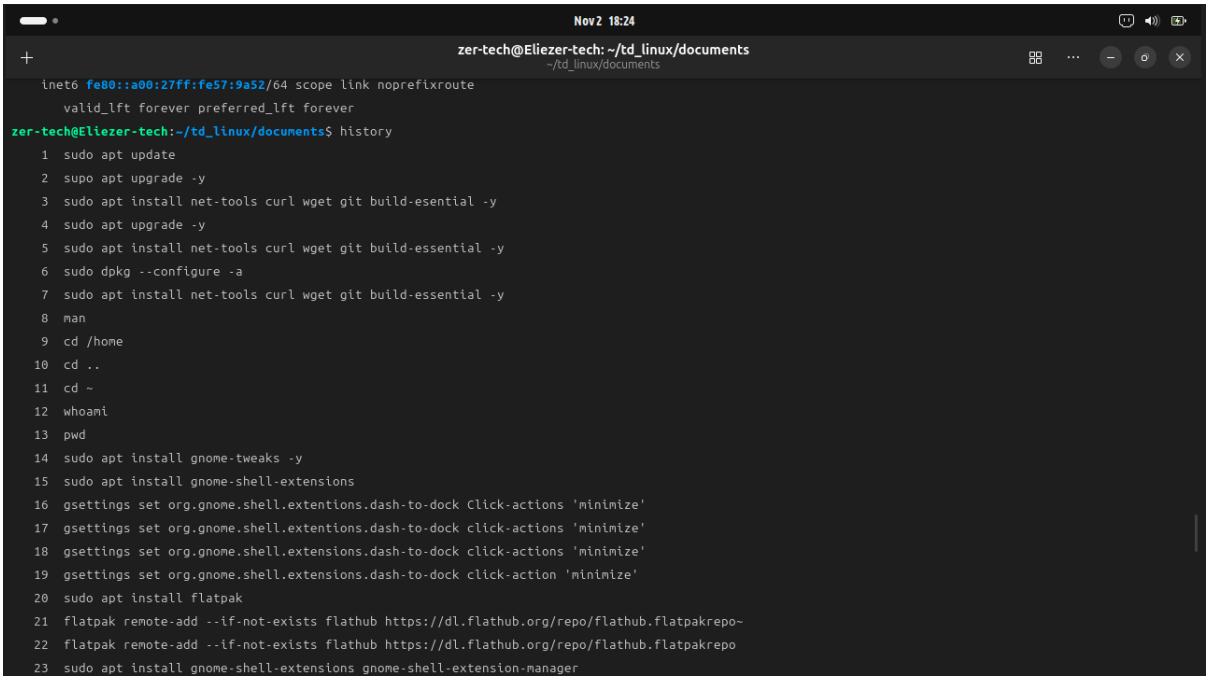
seq 2 2 20 # Affiche les nombres pairs de 2 à 20.

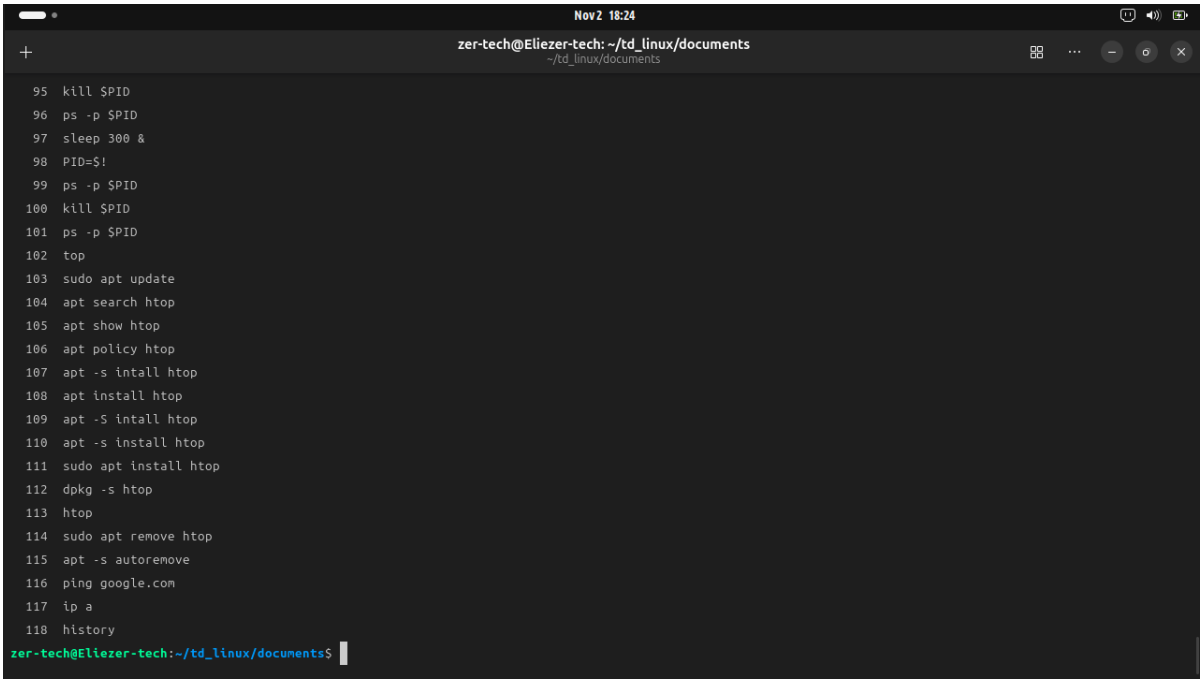
shuf -i 1-100 -n 1 # Affiche un nombre aléatoire entre 1 et 100.

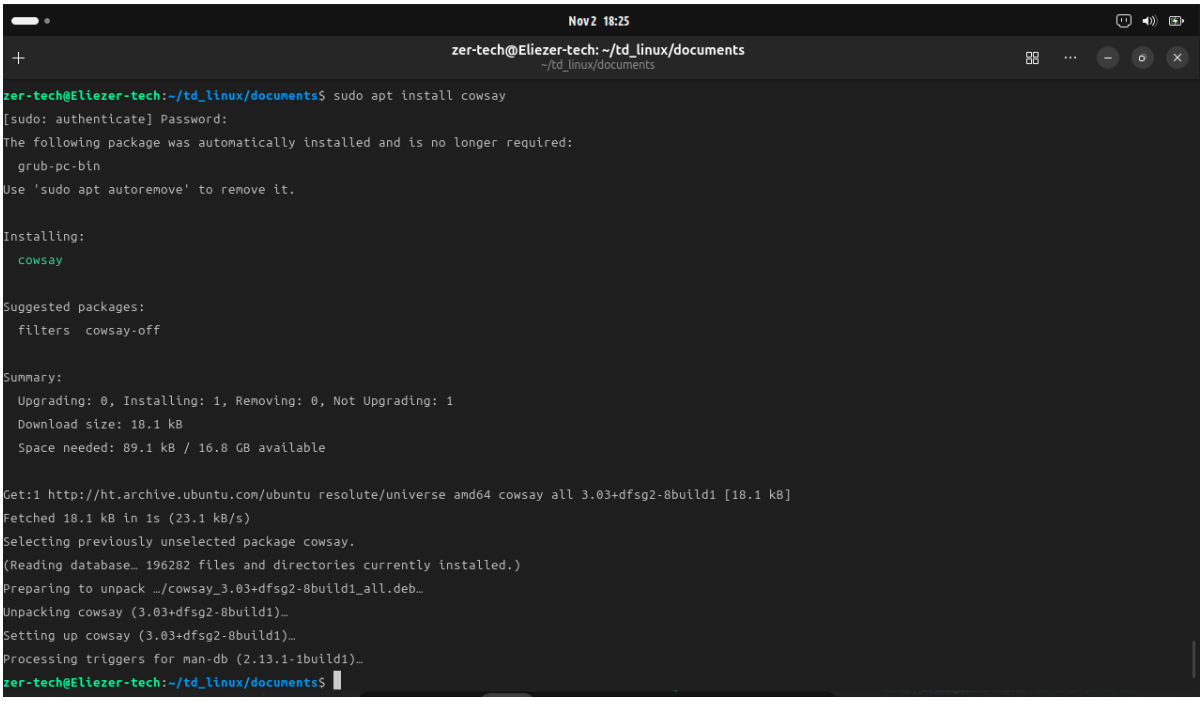

Nov 2 18:18
zer-tech@Eliezer-tech: ~/td_linux/documents — ping google.com
~/td_linux/documents
zer-tech@Eliezer-tech:~/td_linux/documents$ ping google.com
PING google.com (142.251.35.110) 56(84) bytes of data.
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=1 ttl=118 time=24.6 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=2 ttl=118 time=28.1 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=3 ttl=118 time=27.4 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=4 ttl=118 time=22.7 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=5 ttl=118 time=26.7 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=6 ttl=118 time=48.5 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=7 ttl=118 time=52.6 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=8 ttl=118 time=76.7 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=9 ttl=118 time=53.2 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=10 ttl=118 time=42.6 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=11 ttl=118 time=59.0 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=12 ttl=118 time=51.6 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=13 ttl=118 time=57.6 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=14 ttl=118 time=50.5 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=15 ttl=118 time=44.5 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=16 ttl=118 time=52.7 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=17 ttl=118 time=49.4 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=18 ttl=118 time=65.7 ms
Files


Nov 2 18:21
zer-tech@Eliezer-tech: ~/td_linux/documents
~/td_linux/documents
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=289 ttl=118 time=35.6 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=290 ttl=118 time=52.6 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=291 ttl=118 time=66.6 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=292 ttl=118 time=71.8 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=293 ttl=118 time=54.7 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=294 ttl=118 time=70.7 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=295 ttl=118 time=73.1 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=296 ttl=118 time=60.3 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=297 ttl=118 time=64.8 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=298 ttl=118 time=58.7 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=299 ttl=118 time=57.7 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=300 ttl=118 time=76.2 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=301 ttl=118 time=66.4 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=302 ttl=118 time=75.2 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=303 ttl=118 time=63.8 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=304 ttl=118 time=68.8 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=305 ttl=118 time=100 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=306 ttl=118 time=100 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=307 ttl=118 time=114 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=308 ttl=118 time=100 ms
^C
--- google.com ping statistics ---
308 packets transmitted, 303 received, 1.62338% packet loss, time 307932ms
rtt min/avg/max/mdev = 22.215/40.862/132.415/15.681 ms
zer-tech@Eliezer-tech:~/td_linux/documents$


Nov 2 18:22
zer-tech@Eliezer-tech: ~/td_linux/documents
~/td_linux/documents
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=304 ttl=118 time=68.8 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=305 ttl=118 time=100 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=306 ttl=118 time=100 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=307 ttl=118 time=114 ms
64 bytes from lcmiaa-an-in-f14.1e100.net (142.251.35.110): icmp_seq=308 ttl=118 time=100 ms
^C
--- google.com ping statistics ---
308 packets transmitted, 303 received, 1.62338% packet loss, time 307932ms
rtt min/avg/max/mdev = 22.215/40.862/132.415/15.681 ms
zer-tech@Eliezer-tech:~/td_linux/documents$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 08:00:27:57:9a:52 brd ff:ff:ff:ff:ff:ff
inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
valid_lft 76196sec preferred_lft 76196sec
inet6 fd17:625c:f037:2:a00:27ff:fe57:9a52/64 scope global dynamic noprefixroute
valid_lft 86171sec preferred_lft 14171sec
inet6 fe80::a00:27ff:fe57:9a52/64 scope link noprefixroute
valid_lft forever preferred_lft forever
zer-tech@Eliezer-tech:~/td_linux/documents$
Terminal

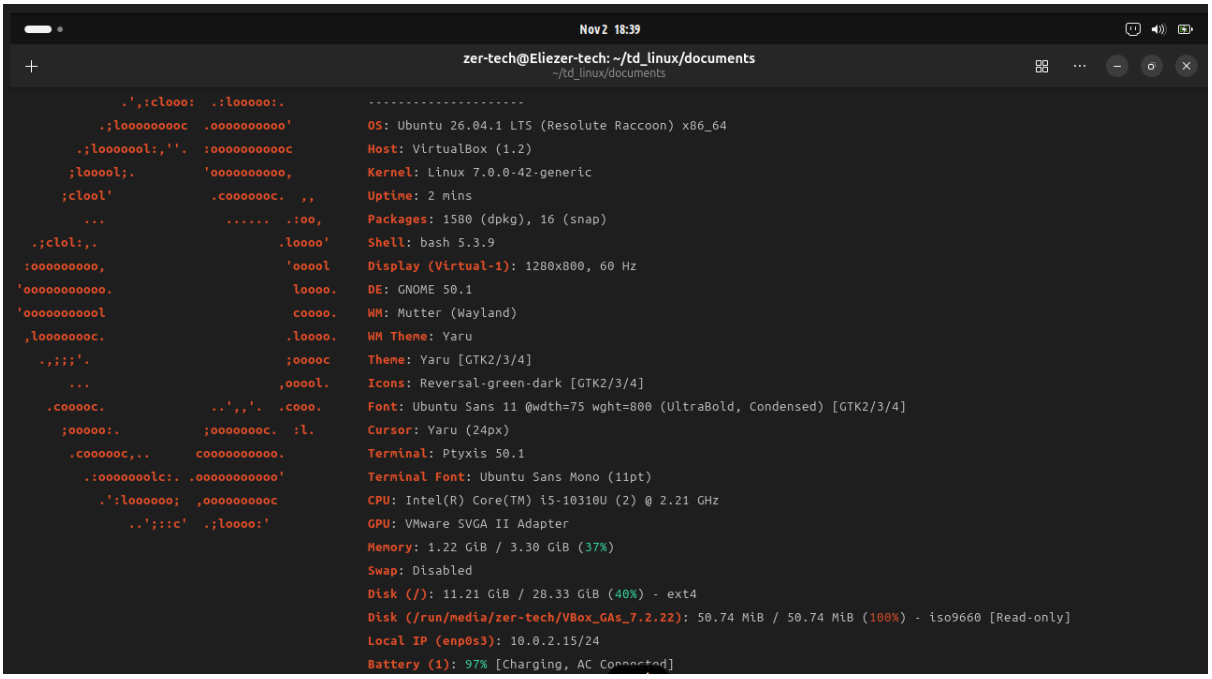

Nov 2 18:24
zer-tech@Eliezer-tech: ~/td_linux/documents
~/td_linux/documents
inet6 fe80::a00:27ff:fe57:9a52/64 scope link noprefixroute
valid_lft forever preferred_lft forever
zer-tech@Eliezer-tech:~/td_linux/documents$ history
1 sudo apt update
2 supo apt upgrade -y
3 sudo apt install net-tools curl wget git build-esential -y
4 sudo apt upgrade -y
5 sudo apt install net-tools curl wget git build-essential -y
6 sudo dpkg --configure -a
7 sudo apt install net-tools curl wget git build-essential -y
8 man
9 cd /home
10 cd ..
11 cd ~
12 whoami
13 pwd
14 sudo apt install gnome-tweaks -y
15 sudo apt install gnome-shell-extensions
16 gsettings set org.gnome.shell.extentions.dash-to-dock Click-actions 'minimize'
17 gsettings set org.gnome.shell.extentions.dash-to-dock click-actions 'minimize'
18 gsettings set org.gnome.shell.extensions.dash-to-dock click-actions 'minimize'
19 gsettings set org.gnome.shell.extensions.dash-to-dock click-action 'minimize'
20 sudo apt install flatpak
21 flatpak remote-add --if-not-exists flathub https://dl.flathub.org/repo/flathub.flatpakrepo~
22 flatpak remote-add --if-not-exists flathub https://dl.flathub.org/repo/flathub.flatpakrepo
23 sudo apt install gnome-shell-extensions gnome-shell-extension-manager

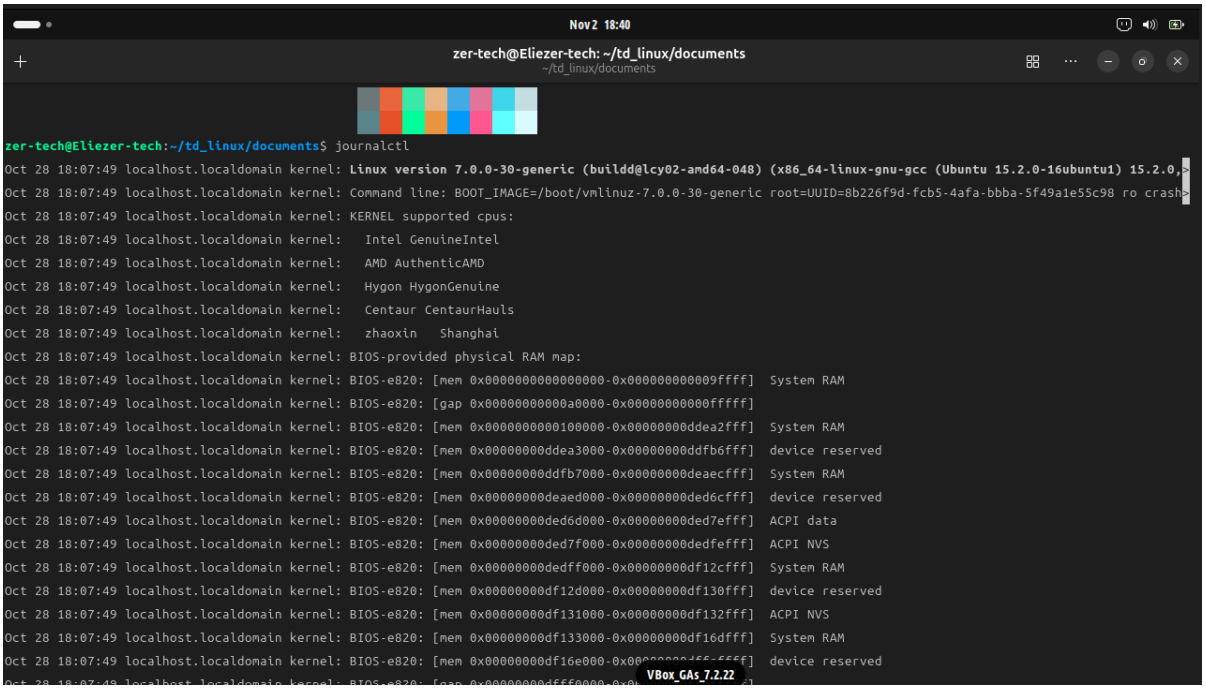

Nov 2 18:24
zer-tech@Eliezer-tech: ~/td_linux/documents
~/td_linux/documents
95 kill $PID
96 ps -p $PID
97 sleep 300 &
98 PID=$!
99 ps -p $PID
100 kill $PID
101 ps -p $PID
102 top
103 sudo apt update
104 apt search htop
105 apt show htop
106 apt policy htop
107 apt -s intall htop
108 apt install htop
109 apt -S intall htop
110 apt -s install htop
111 sudo apt install htop
112 dpkg -s htop
113 htop
114 sudo apt remove htop
115 apt -s autoremove
116 ping google.com
117 ip a
118 history
zer-tech@Eliezer-tech:~/td_linux/documents$

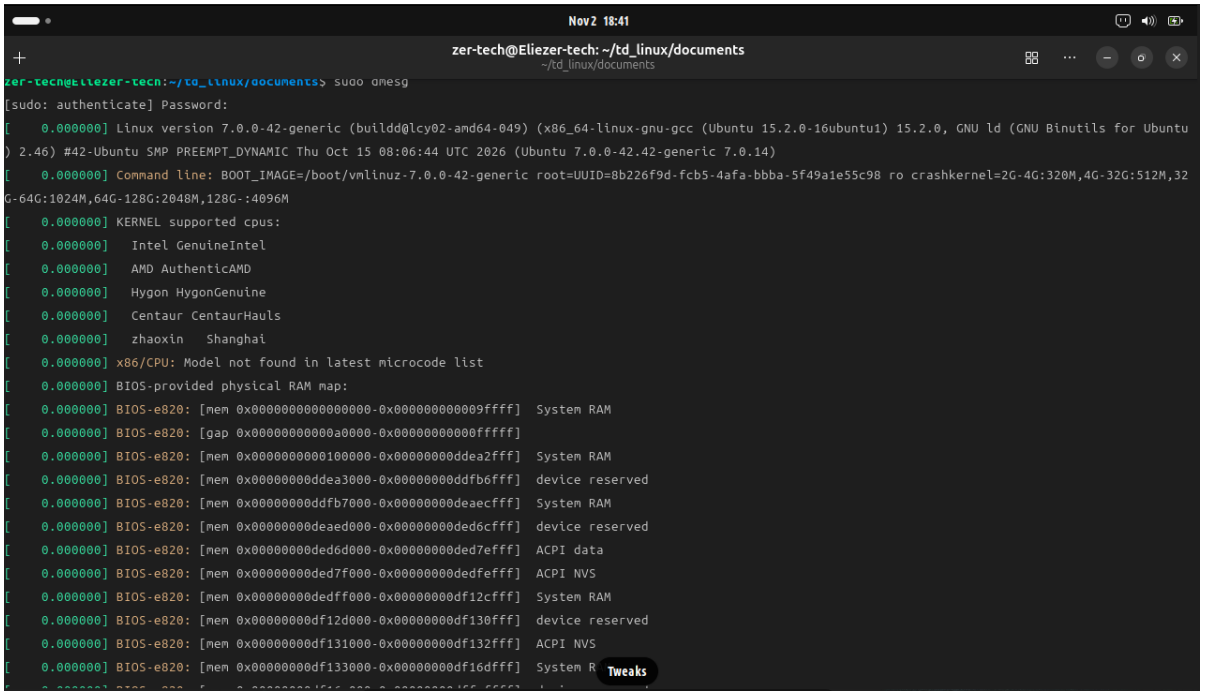

Nov 2 18:25
zer-tech@Eliezer-tech: ~/td_linux/documents
~/td_linux/documents
zer-tech@Eliezer-tech:~/td_linux/documents$ sudo apt install cowsay
[sudo: authenticate] Password:
The following package was automatically installed and is no longer required:
grub-pc-bin
Use 'sudo apt autoremove' to remove it.
Installing:
cowsay
Suggested packages:
filters cowsay-off
Summary:
Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1
Download size: 18.1 kB
Space needed: 89.1 kB / 16.8 GB available
Get:1 http://ht.archive.ubuntu.com/ubuntu resolute/universe amd64 cowsay all 3.03+dfsg2-8build1 [18.1 kB]
Fetched 18.1 kB in 1s (23.1 kB/s)
Selecting previously unselected package cowsay.
(Reading database… 196282 files and directories currently installed.)
Preparing to unpack …/cowsay_3.03+dfsg2-8build1_all.deb…
Unpacking cowsay (3.03+dfsg2-8build1)…
Setting up cowsay (3.03+dfsg2-8build1)…
Processing triggers for man-db (2.13.1-1build1)…
zer-tech@Eliezer-tech:~/td_linux/documents$

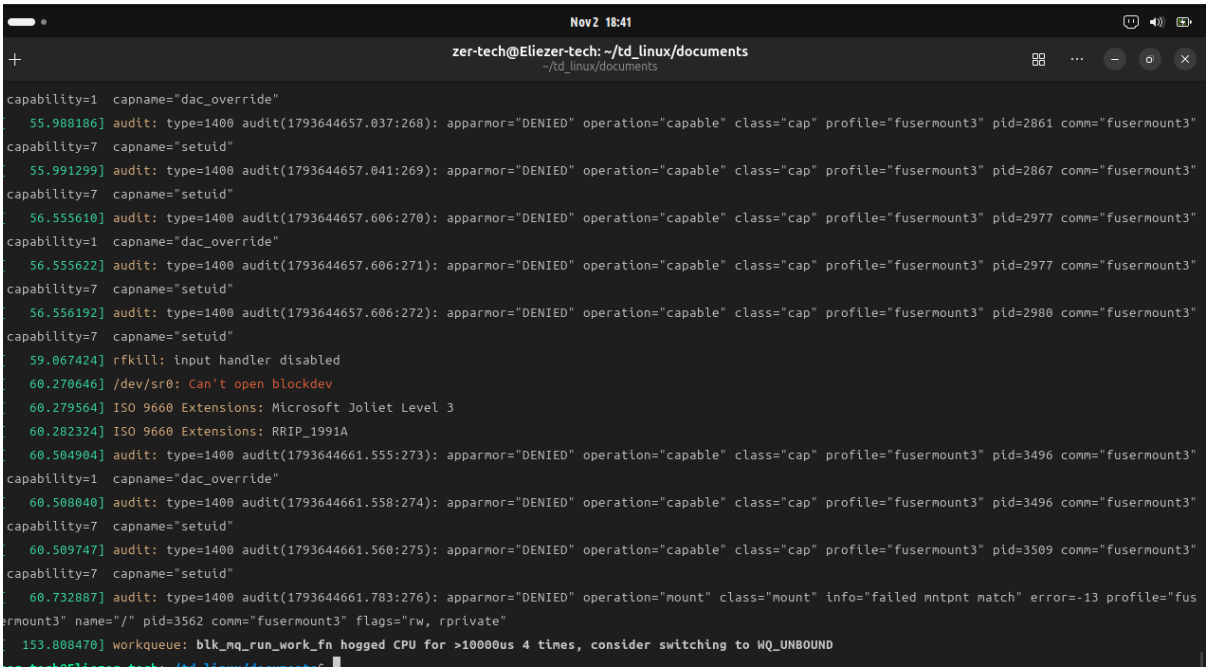


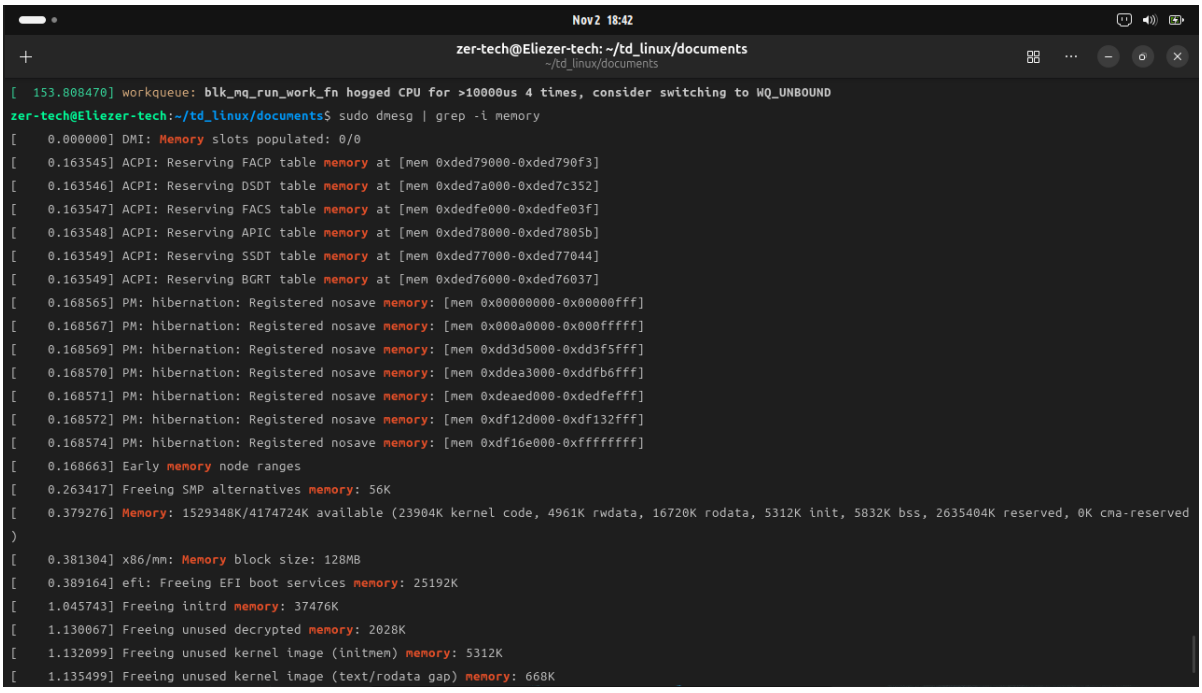
Nous avons utilisé Fastfetch au lieu de Neofetch car le packet était absent

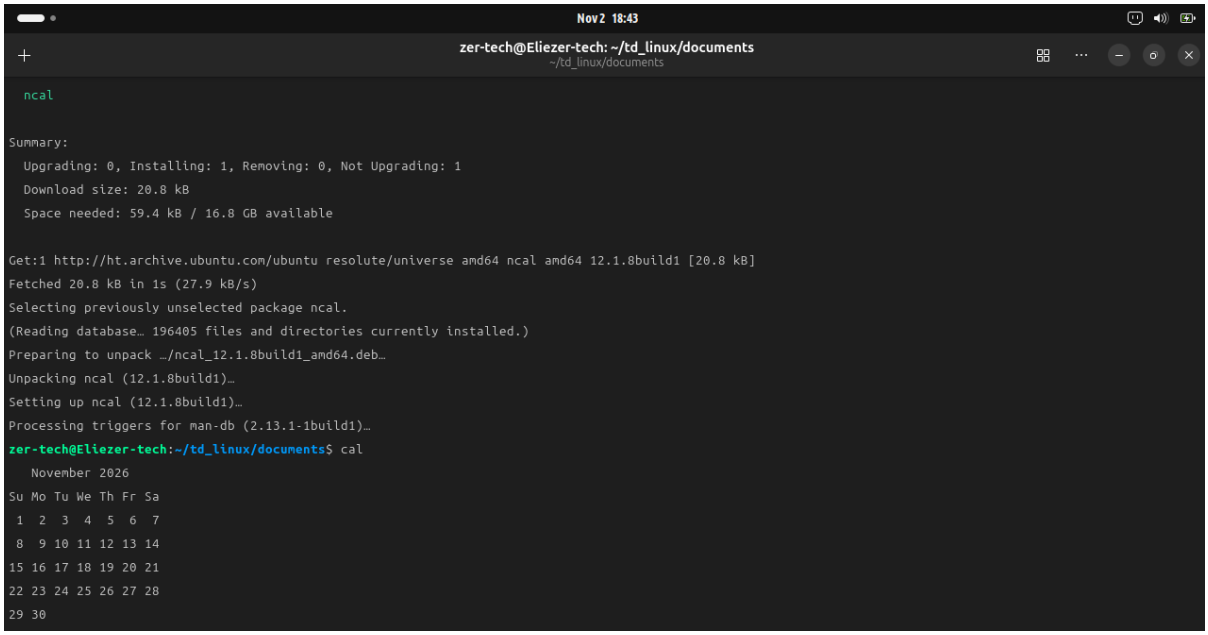


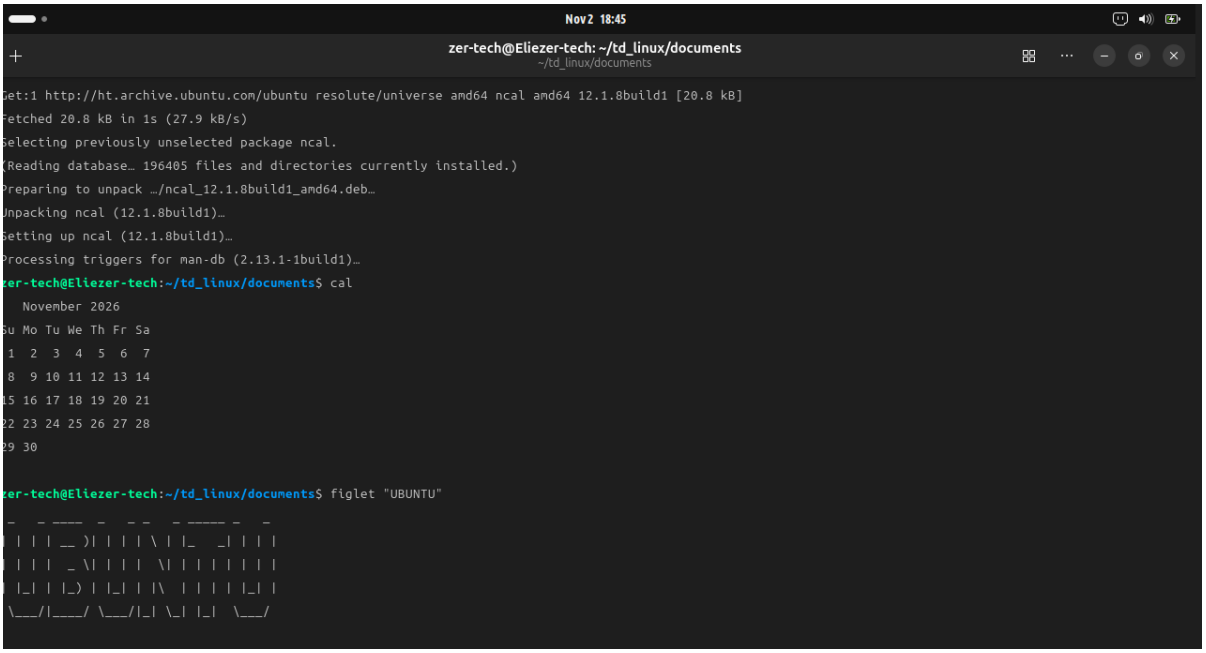

Nov 2 18:40
zer-tech@Eliezer-tech: ~/td_linux/documents
~/td_linux/documents
zer-tech@Eliezer-tech:~/td_linux/documents$ journalctl
Oct 28 18:07:49 localhost.localdomain kernel: Linux version 7.0.0-30-generic (buildd@lcy02-amd64-048) (x86_64-linux-gnu-gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0,
Oct 28 18:07:49 localhost.localdomain kernel: Command line: BOOT_IMAGE=/boot/vmlinuz-7.0.0-30-generic root=UUID=8b226f9d-fcb5-4afa-bbba-5f49a1e55c98 ro crash
Oct 28 18:07:49 localhost.localdomain kernel: KERNEL supported cpus:
Oct 28 18:07:49 localhost.localdomain kernel: Intel GenuineIntel
Oct 28 18:07:49 localhost.localdomain kernel: AMD AuthenticAMD
Oct 28 18:07:49 localhost.localdomain kernel: Hygon HygonGenuine
Oct 28 18:07:49 localhost.localdomain kernel: Centaur CentaurHauls
Oct 28 18:07:49 localhost.localdomain kernel: zhaoxin Shanghai
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-provided physical RAM map:
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x0000000000000000-0x000000000009ffff] System RAM
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [gap 0x00000000000a0000-0x00000000000fffff]
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x0000000000100000-0x00000000ddea2fff] System RAM
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000ddea3000-0x00000000ddfb6fff] device reserved
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000ddfb7000-0x00000000deaecfff] System RAM
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000deaed000-0x00000000ded6cfff] device reserved
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000ded6d000-0x00000000ded7efff] ACPI data
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000ded7f000-0x00000000dedfefff] ACPI NVS
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000dedff000-0x00000000df12cfff] System RAM
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000df12d000-0x00000000df130fff] device reserved
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000df131000-0x00000000df132fff] ACPI NVS
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000df133000-0x00000000df16dfff] System RAM
Oct 28 18:07:49 localhost.localdomain kernel: BIOS-e820: [mem 0x00000000df16e000-0x
VBox_GAs_7.2.22

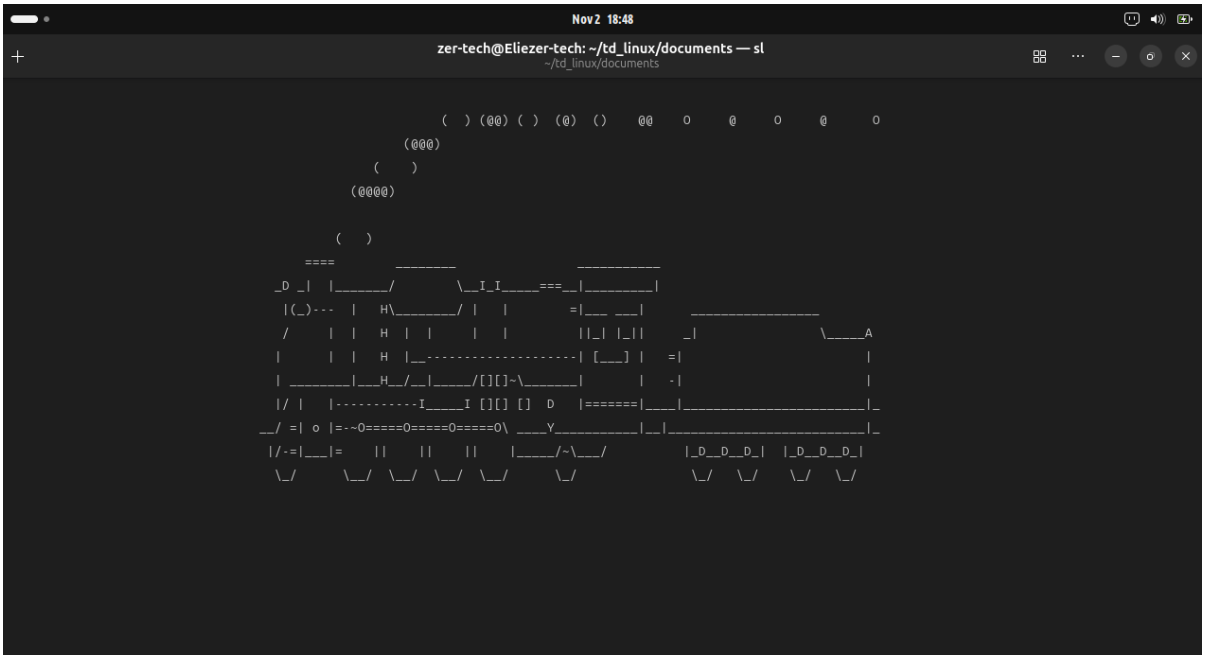

Nov 2 18:41
zer-tech@Eliezer-tech: ~/td_linux/documents
~/td_linux/documents
zer-tech@Eliezer-tech:~/td_linux/documents$ sudo dmesg
[sudo: authenticate] Password:
[0.000000] Linux version 7.0.0-42-generic (buildd@lcy02-amd64-049) (x86_64-linux-gnu-gcc (Ubuntu 15.2.0-16ubuntu1) 15.2.0, GNU ld (GNU Binutils for Ubuntu) 2.46) #42-Ubuntu SMP PREEMPT_DYNAMIC Thu Oct 15 08:06:44 UTC 2026 (Ubuntu 7.0.0-42.42-generic 7.0.14)
[0.000000] Command line: BOOT_IMAGE=/boot/vmlinuz-7.0.0-42-generic root=UUID=8b226f9d-fcb5-4afa-bbba-5f49a1e55c98 ro crashkernel=2G-4G:320M,4G-32G:512M,32G-64G:1024M,64G-128G:2048M,128G-:4096M
[0.000000] KERNEL supported cpus:
[0.000000] Intel GenuineIntel
[0.000000] AMD AuthenticAMD
[0.000000] Hygon HygonGenuine
[0.000000] Centaur CentaurHauls
[0.000000] zhaoxin Shanghai
[0.000000] x86/CPU: Model not found in latest microcode list
[0.000000] BIOS-provided physical RAM map:
[0.000000] BIOS-e820: [mem 0x0000000000000000-0x000000000009ffff] System RAM
[0.000000] BIOS-e820: [gap 0x00000000000a0000-0x00000000000fffff]
[0.000000] BIOS-e820: [mem 0x0000000000100000-0x00000000ddea2fff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000ddea3000-0x00000000ddfb6fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000ddfb7000-0x00000000deaecfff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000deaed000-0x00000000ded6cfff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000ded6d000-0x00000000ded7efff] ACPI data
[0.000000] BIOS-e820: [mem 0x00000000ded7f000-0x00000000dedfefff] ACPI NVS
[0.000000] BIOS-e820: [mem 0x00000000dedff000-0x00000000df12cfff] System RAM
[0.000000] BIOS-e820: [mem 0x00000000df12d000-0x00000000df130fff] device reserved
[0.000000] BIOS-e820: [mem 0x00000000df131000-0x00000000df132fff] ACPI NVS
[0.000000] BIOS-e820: [mem 0x00000000df133000-0x00000000df16dfff] System R
Tweaks

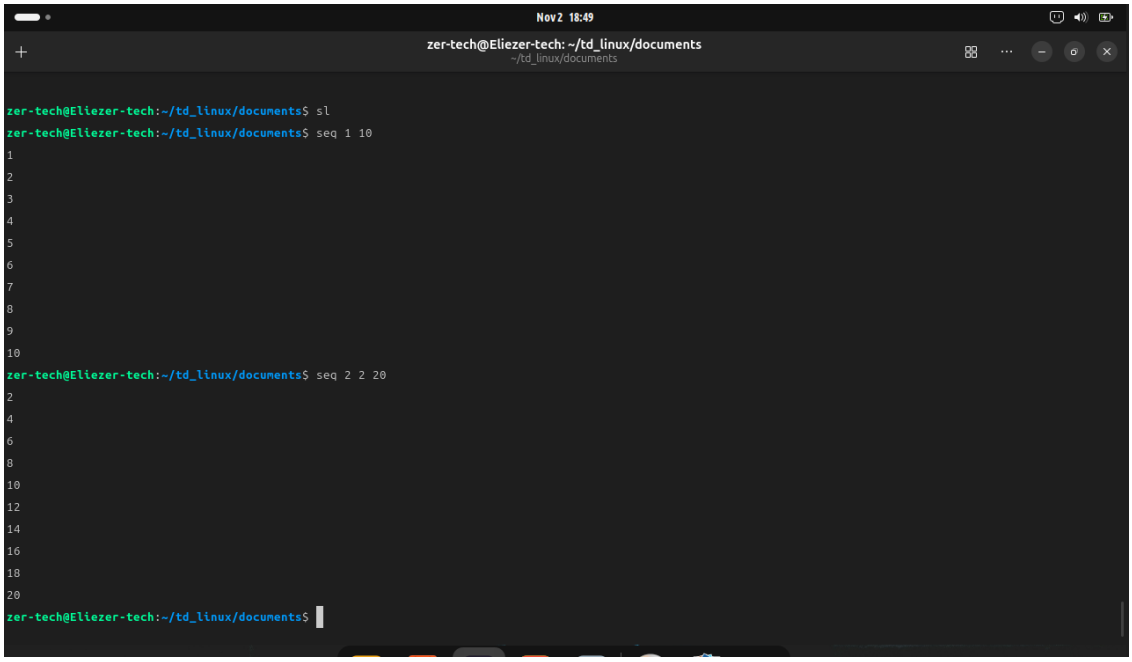

Nov 2 18:41
zer-tech@Eliezer-tech: ~/td_linux/documents
~/td_linux/documents
capability=1 capname="dac_override"
[55.988186] audit: type=1400 audit(1793644657.037:268): apparmor="DENIED" operation="capable" class="cap" profile="fusermount3" pid=2861 comm="fusermount3"
capability=7 capname="setuid"
[55.991299] audit: type=1400 audit(1793644657.041:269): apparmor="DENIED" operation="capable" class="cap" profile="fusermount3" pid=2867 comm="fusermount3"
capability=7 capname="setuid"
[56.555610] audit: type=1400 audit(1793644657.606:270): apparmor="DENIED" operation="capable" class="cap" profile="fusermount3" pid=2977 comm="fusermount3"
capability=1 capname="dac_override"
[56.555622] audit: type=1400 audit(1793644657.606:271): apparmor="DENIED" operation="capable" class="cap" profile="fusermount3" pid=2977 comm="fusermount3"
capability=7 capname="setuid"
[56.556192] audit: type=1400 audit(1793644657.606:272): apparmor="DENIED" operation="capable" class="cap" profile="fusermount3" pid=2980 comm="fusermount3"
capability=7 capname="setuid"
[59.067424] rfkill: input handler disabled
[60.270646] /dev/sr0: Can't open blockdev
[60.279564] ISO 9660 Extensions: Microsoft Joliet Level 3
[60.282324] ISO 9660 Extensions: RRIP_1991A
[60.504904] audit: type=1400 audit(1793644661.555:273): apparmor="DENIED" operation="capable" class="cap" profile="fusermount3" pid=3496 comm="fusermount3"
capability=1 capname="dac_override"
[60.508040] audit: type=1400 audit(1793644661.558:274): apparmor="DENIED" operation="capable" class="cap" profile="fusermount3" pid=3496 comm="fusermount3"
capability=7 capname="setuid"
[60.509747] audit: type=1400 audit(1793644661.560:275): apparmor="DENIED" operation="capable" class="cap" profile="fusermount3" pid=3509 comm="fusermount3"
capability=7 capname="setuid"
[60.732887] audit: type=1400 audit(1793644661.783:276): apparmor="DENIED" operation="mount" class="mount" info="failed mntpnt match" error=-13 profile="fus
rmount3" name="/" pid=3562 comm="fusermount3" flags="rw, rprivate"
[153.808470] workqueue: blk_mq_run_work_fn hogged CPU for >10000us 4 times, consider switching to WQ_UNBOUND

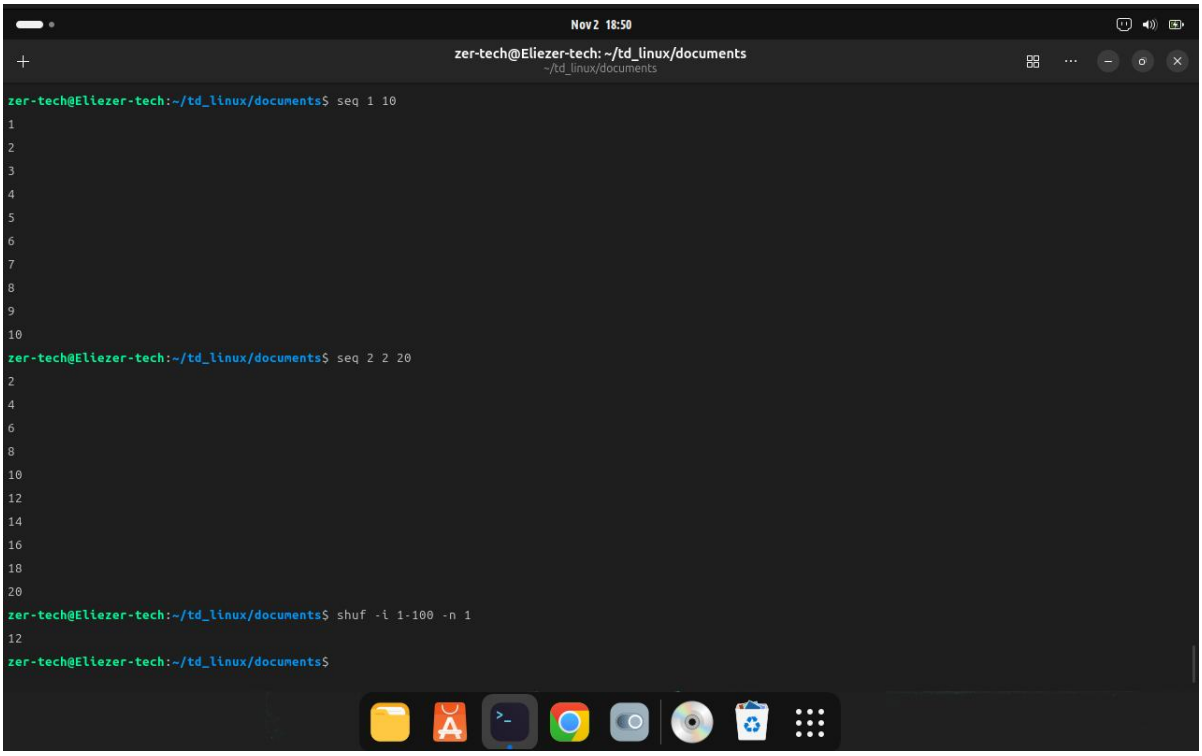

Nov 2 18:42
zer-tech@Eliezer-tech: ~/td_linux/documents
~/td_linux/documents
[153.808470] workqueue: blk_mq_run_work_fn hogged CPU for >10000us 4 times, consider switching to WQ_UNBOUND
zer-tech@Eliezer-tech:~/td_linux/documents$ sudo dmesg | grep -i memory
[0.000000] DMI: Memory slots populated: 0/0
[0.163545] ACPI: Reserving FACP table memory at [mem 0xded79000-0xded790f3]
[0.163546] ACPI: Reserving DSDT table memory at [mem 0xded7a000-0xded7c352]
[0.163547] ACPI: Reserving FACS table memory at [mem 0xdedfe000-0xdedfe03f]
[0.163548] ACPI: Reserving APIC table memory at [mem 0xded78000-0xded7805b]
[0.163549] ACPI: Reserving SSDT table memory at [mem 0xded77000-0xded77044]
[0.163549] ACPI: Reserving BGRT table memory at [mem 0xded76000-0xded76037]
[0.168565] PM: hibernation: Registered nosave memory: [mem 0x00000000-0x00000fff]
[0.168567] PM: hibernation: Registered nosave memory: [mem 0x000a0000-0x000fffff]
[0.168569] PM: hibernation: Registered nosave memory: [mem 0xdd3d5000-0xdd3f5fff]
[0.168570] PM: hibernation: Registered nosave memory: [mem 0xddea3000-0xddfb6fff]
[0.168571] PM: hibernation: Registered nosave memory: [mem 0xdeaed000-0xdedfefff]
[0.168572] PM: hibernation: Registered nosave memory: [mem 0xdf12d000-0xdf132fff]
[0.168574] PM: hibernation: Registered nosave memory: [mem 0xdf16e000-0xffffffff]
[0.168663] Early memory node ranges
[0.263417] Freeing SMP alternatives memory: 56K
[0.379276] Memory: 1529348K/4174724K available (23904K kernel code, 4961K rwdata, 16720K rodata, 5312K init, 5832K bss, 2635404K reserved, 0K cma-reserved
)
[0.381304] x86/mm: Memory block size: 128MB
[0.389164] efi: Freeing EFI boot services memory: 25192K
[1.045743] Freeing initrd memory: 37476K
[1.130067] Freeing unused decrypted memory: 2028K
[1.132099] Freeing unused kernel image (initmem) memory: 5312K
[1.135499] Freeing unused kernel image (text/rodata gap) memory: 668K


Nov 2 18:43
zer-tech@Eliezer-tech: ~/td_linux/documents
~/td_linux/documents
ncal
Summary:
Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 1
Download size: 20.8 kB
Space needed: 59.4 kB / 16.8 GB available
Get:1 http://ht.archive.ubuntu.com/ubuntu resolute/universe amd64 ncal amd64 12.1.8build1 [20.8 kB]
Fetched 20.8 kB in 1s (27.9 kB/s)
Selecting previously unselected package ncal.
(Reading database… 196405 files and directories currently installed.)
Preparing to unpack …/ncal_12.1.8build1_amd64.deb…
Unpacking ncal (12.1.8build1)…
Setting up ncal (12.1.8build1)…
Processing triggers for man-db (2.13.1-1build1)…
zer-tech@Eliezer-tech:~/td_linux/documents$ cal
November 2026
Su Mo Tu We Th Fr Sa
1 2 3 4 5 6 7
8 9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30


Nov 2 18:45
zer-tech@Eliezer-tech: ~/td_linux/documents
~/td_linux/documents
zer-tech@Eliezer-tech:~/td_linux/documents$ cal
November 2026
Su Mo Tu We Th Fr Sa
1 2 3 4 5 6 7
8 9 10 11 12 13 14
15 16 17 18 19 20 21
22 23 24 25 26 27 28
29 30
zer-tech@Eliezer-tech:~/td_linux/documents$ figlet "UBUNTU"


Nov 2 18:48
zer-tech@Eliezer-tech: ~/td_linux/documents — sl
~/td_linux/documents


Nov 2 18:49
zer-tech@Eliezer-tech: ~/td_linux/documents
~/td_linux/documents
zer-tech@Eliezer-tech:~/td_linux/documents$ sl
zer-tech@Eliezer-tech:~/td_linux/documents$ seq 1 10
1
2
3
4
5
6
7
8
9
10
zer-tech@Eliezer-tech:~/td_linux/documents$ seq 2 2 20
2
4
6
8
10
12
14
16
18
20
zer-tech@Eliezer-tech:~/td_linux/documents$


Nov 2 18:50
zer-tech@Eliezer-tech: ~/td_linux/documents
~/td_linux/documents
zer-tech@Eliezer-tech:~/td_linux/documents$ seq 1 10
1
2
3
4
5
6
7
8
9
10
zer-tech@Eliezer-tech:~/td_linux/documents$ seq 2 2 20
2
4
6
8
10
12
14
16
18
20
zer-tech@Eliezer-tech:~/td_linux/documents$ shuf -i 1-100 -n 1
12
zer-tech@Eliezer-tech:~/td_linux/documents$

EXERCICE 5

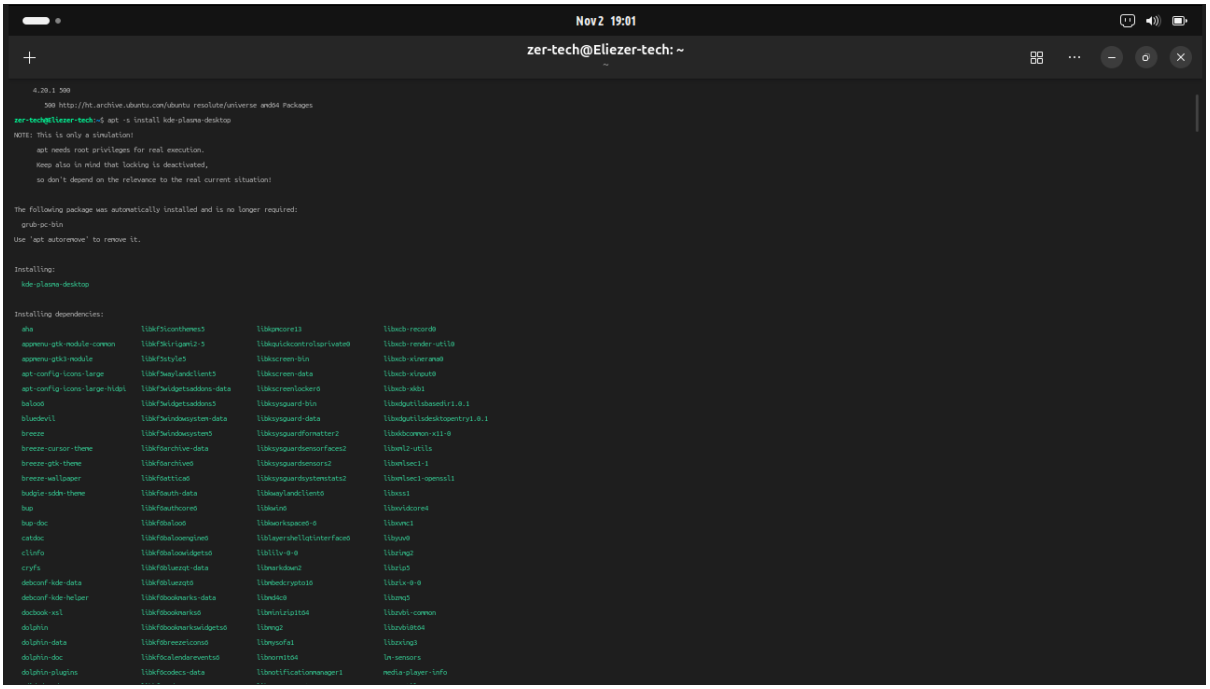
Cette expérience vise à installer, configurer et comparer deux environnements de bureau distincts (KDE Plasma et XFCE) afin d'analyser leur comportement et leur impact sur le système.

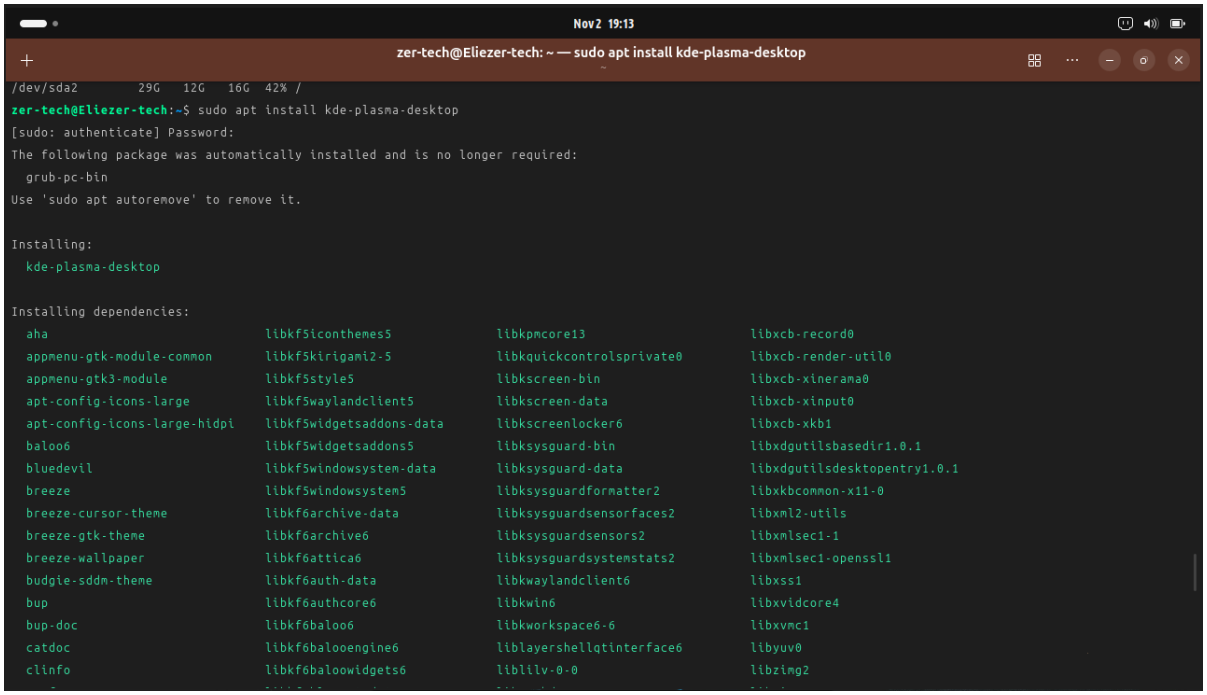
1. Identification de l'environnement de bureau actif et du type de session graphique (serveur d'affichage) via l'analyse des variables d'environnement du système.
2. Vérification de la disponibilité et du statut des paquets logiciels requis dans le gestionnaire de paquets de la distribution.
3. Simulation des installations pour évaluer en amont l'espace disque nécessaire et examiner l'arbre des dépendances logicielles de chaque interface.
4. Installation effective des deux environnements avec les privilèges d'administration, en prenant soin de conserver le gestionnaire de connexion d'origine en cas d'invite système.
5. Déconnexion de la session courante après sauvegarde des travaux, puis tests successifs d'ouverture de session sous Plasma, puis sous XFCE.

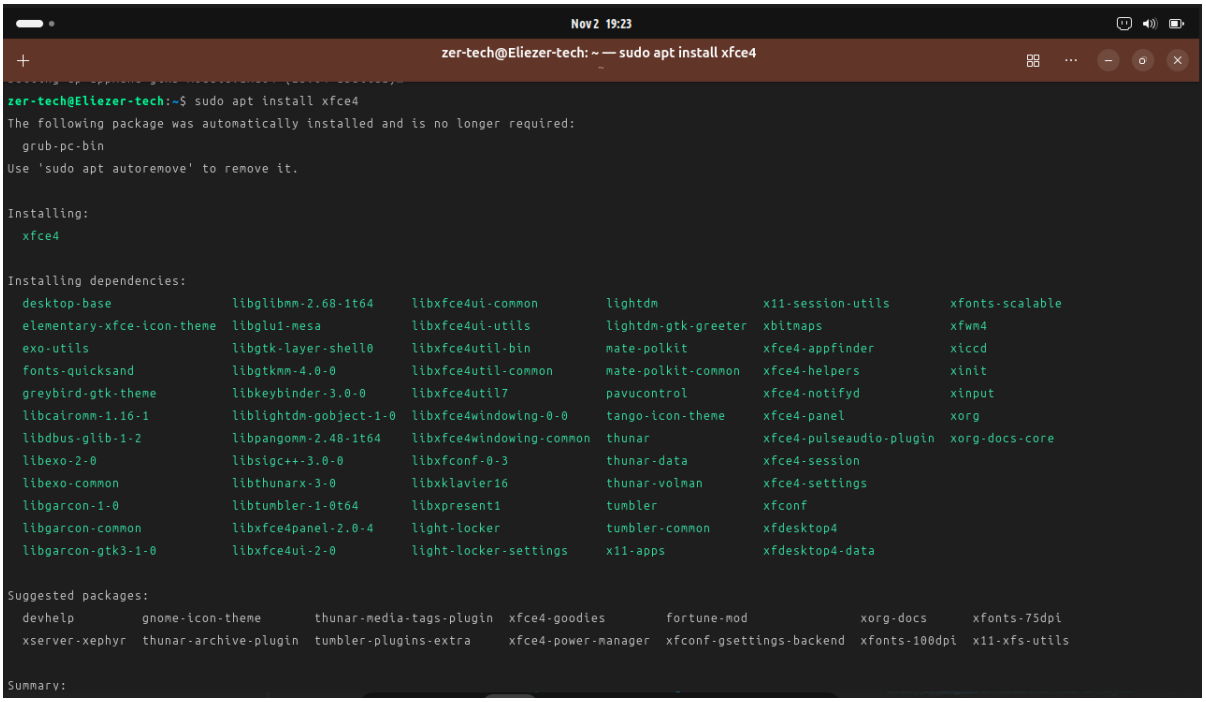
Évaluation et comparaison

Pour chaque interface, un banc d'essai standardisé est appliqué afin de tester l'ergonomie et la configuration :

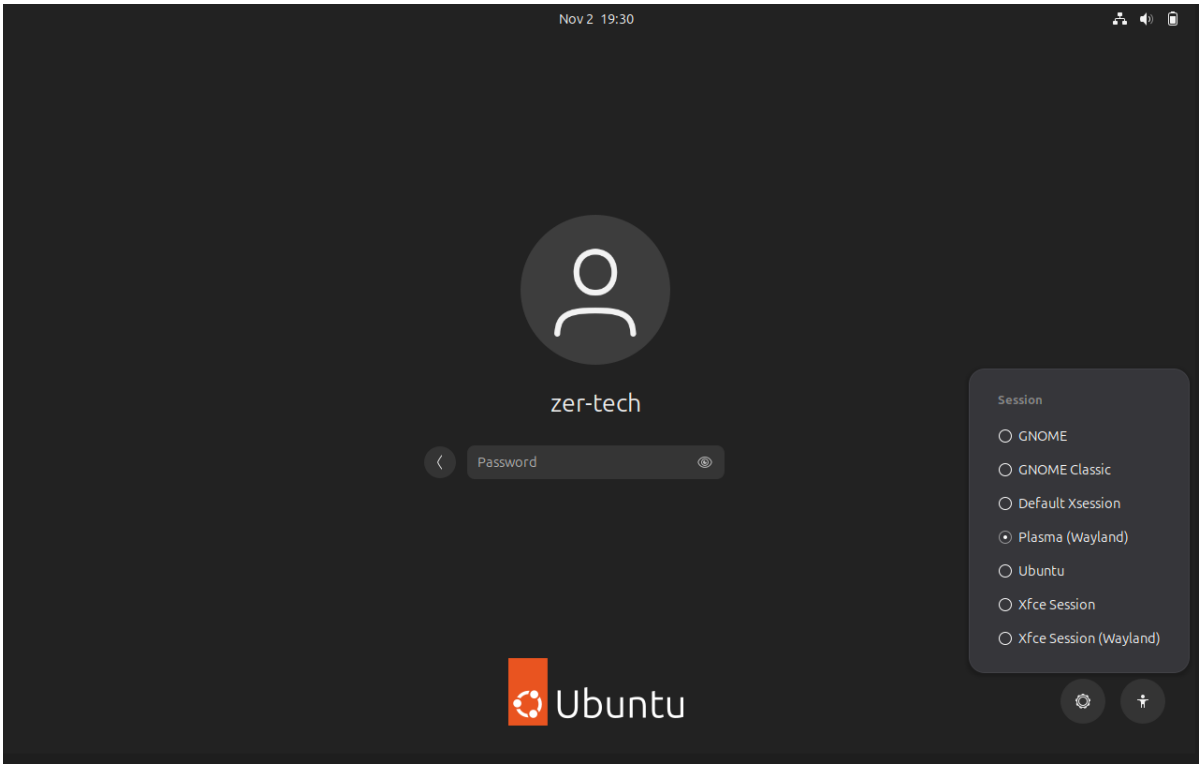
- L'exécution des outils de base (terminal, gestionnaire de fichiers).
- L'accès aux documents de travail.
- La personnalisation de l'interface (fond d'écran, création de raccourcis).
- L'exploration du panneau d'administration des paramètres système.

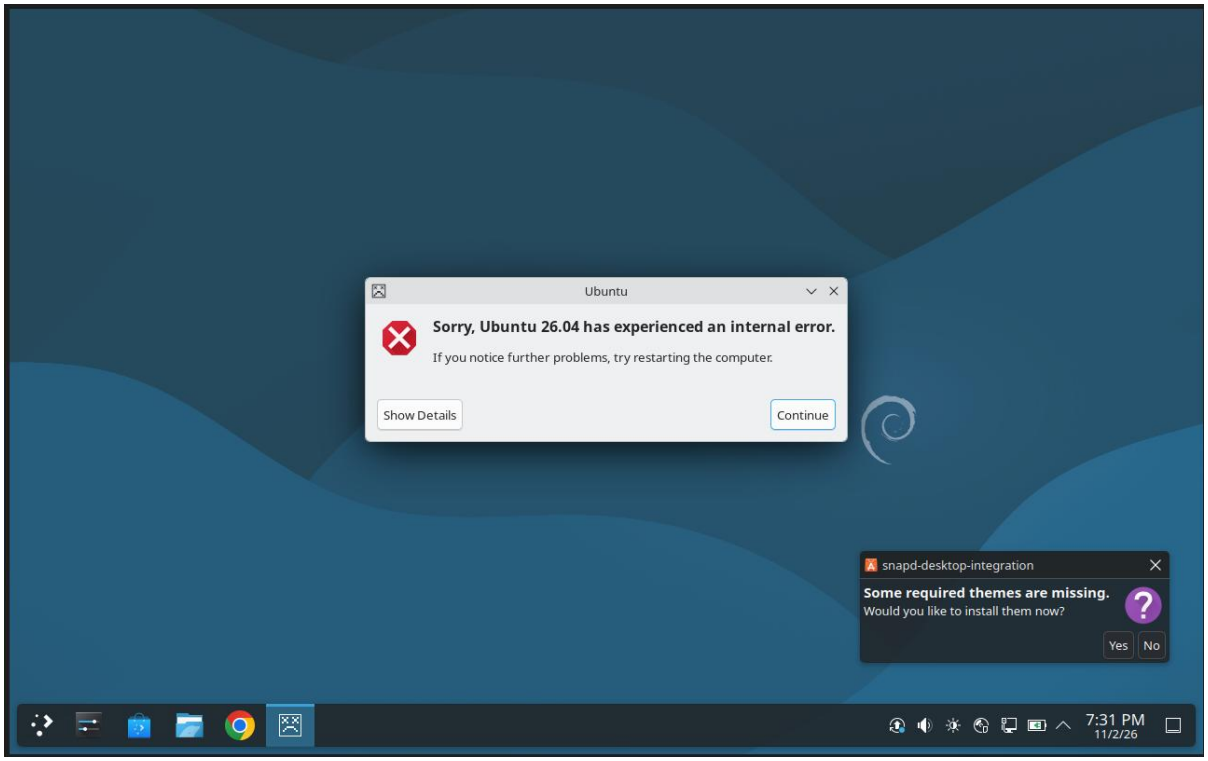

Nov 2 19:01
zer-tech@Eliezer-tech: ~
4.20.1 500
500 http://nt.archive.ubuntu.com/ubuntu resolute/universe amd64 Packages
zer-tech@Eliezer-tech:~$ apt -s install kde-plasma-desktop
NOTE: This is only a simulation!
apt needs root privileges for real execution.
Keep also in mind that locking is deactivated,
so don't depend on the relevance to the real current situation!
The following package was automatically installed and is no longer required:
grub-pc-bin
Use 'apt autoremove' to remove it.
Installing:
kde-plasma-desktop
Installing dependencies:
aha
appmenu-gtk-module-common
appmenu-gtk3-module
apt-config-icons-large
apt-config-icons-large-hidpi
baloo6
bluedevil
breeze
breeze-cursor-theme
breeze-gtk-theme
breeze-wallpaper
budgie-sddm-theme
bup
bup-doc
catdoc
clinfo
cryfs
debconf-kde-data
debconf-kde-helper
docbook-xsl
dolphin
dolphin-data
dolphin-doc
dolphin-plugins
libkf5iconthemes5
libkf5kirigami2-5
libkf5style5
libkf5waylandclient5
libkf5widgetsaddons-data
libkf5widgetsaddons5
libkf5windowsystem-data
libkf5windowsystem5
libkf6archive-data
libkf6archive6
libkf6attica6
libkf6auth-data
libkf6authcore6
libkf6baloo6
libkf6balooengine6
libkf6bookmarks-data
libkf6bookmarks6
libkf6bookmarkswidgets6
libkf6breezeicons6
libkf6codecs-data
libkf6calendarevents6
libkf6coreaddons-data
libkf6coreaddons6
libkf6crash6
libkf6dbusaddons-data
libkf6dbusaddons6
libkf6filemetadata-data
libkf6filemetadata6
libkf6globalaccel-data
libkf6globalaccel6
libkf6guiaddons6
libkf6i18n-data
libkf6i18n6
libkf6iconthemes-data
libkf6iconthemes6
libkf6idletime6
libkf6itemmodels6
libkf6itemviews6
libkf6jobwidgets-data
libkf6jobwidgets6
libkf6kcmutils-data
libkf6kcmutils6
libkf6kio-data
libkf6kio6
libkf6kirigami6
libkf6kirigamiaddons6
libkf6kirigamiaddons6-data
libkf6kirigamiplatform6
libkf6newstuff-data
libkf6newstuff6
libkf6notifications-data
libkf6notifications6
libkf6package-data
libkf6package6
libkf6parts-data
libkf6parts6
libkf6pty-data
libkf6pty6
libkf6purpose-bin
libkf6purpose-data
libkf6purpose6
libkf6quickcharts-data
libkf6quickcharts6
libkf6runner6
libkf6service-bin
libkf6service-data
libkf6service6
libkf6solid-data
libkf6solid6
libkf6sonnet-data
libkf6sonnet6
libkf6syntaxhighlighting-data
libkf6syntaxhighlighting6
libkf6texteditor-data
libkf6texteditor6
libkf6textwidgets-data
libkf6textwidgets6
libkf6unitconversion-data
libkf6unitconversion6
libkf6widgetsaddons-data
libkf6widgetsaddons6
libkf6windowsystem-data
libkf6windowsystem6
libkf6xmlgui-data
libkf6xmlgui6
libkgantt2-6
libkgeomap6
libkgeomap6-data
libkgpg6
libkgpg6-data
libkholidays6
libkholidays6-data
libkmahjongg6
libkmahjongg6-data
libkmbox6
libkmbox6-data
libkmime6
libkmime6-data
libkpimcalendaring6
libkpimcalendaring6-data
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libkpimcommon6-data
libkpimgapi6
libkpimgapi6-data
libkpimgmail6
libkpimgmail6-data
libkpimgravatar6
libkpimgravatar6-data
libkpimidentities6
libkpimidentities6-data
libkpimimportwizard6
libkpimimportwizard6-data
libkpimmailcommon6
libkpimmailcommon6-data
libkpimmailimporter6
libkpimmailimporter6-data
libkpimmailtransport6
libkpimmailtransport6-data
libkpimmbox6
libkpimmbox6-data
libkpimmessagecomposer6
libkpimmessagecomposer6-data
libkpimmessagecore6
libkpimmessagecore6-data
libkpimmessagelist6
libkpimmessagelist6-data
libkpimmessageviewer6
libkpimmessageviewer6-data
libkpimmimetreeparser6
libkpimmimetreeparser6-data
libkpimndbussupport6
libkpimndbussupport6-data
libkpimnotes6
libkpimnotes6-data
libkpimpgp6
libkpimpgp6-data
libkpimsearch6
libkpimsearch6-data
libkpimsmtp6
libkpimsmtp6-data
libkpimtemplate6
libkpimtemplate6-data
libkpimtextedit6
libkpimtextedit6-data
libkpimtnef6
libkpimtnef6-data
libkpimxmlrpc6
libkpimxmlrpc6-data
libkpmcore13
libkquickcontrolsprivate6
libkscreen-bin
libkscreen-data
libkscreenlocker6
libksysguard-bin
libksysguard-data
libksysguardformatter2
libksysguardsensorfaces2
libksysguardsensors2
libksysguardsystemstats2
libkwaylandclient6
libkwin6
libkworkspace6-6
liblayershellqtinterface6
liblily-0-0
libmarkdown2
libmbedcrypto16
libmd4c0
libminizip1t64
libmng2
libmysofa1
libnormit64
libnotificationmanager1
libnss3
libnvidcore4
libnvnc1
libnvui9
libqrtr-glib0
libqt6waylandcompositor6
libqt6waylandeglclienthwintegration6
libqt6waylandeglcompositorhwintegration6
libqt6waylandeglcompositorhwintegration-plugin
libqt6waylandeglcompositorhwintegration-plugin-plugins
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libqt6waylandeglcompositorhwintegration-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin
libqt6waylandeglcompositorhwintegration-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins
libqt6waylandeglcompositorhwintegration-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin
libqt6waylandeglcompositorhwintegration-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins
libqt6waylandeglcompositorhwintegration-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin
libqt6waylandeglcompositorhwintegration-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins
libqt6waylandeglcompositorhwintegration-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin
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libqt6waylandeglcompositorhwintegration-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin-plugins-plugin

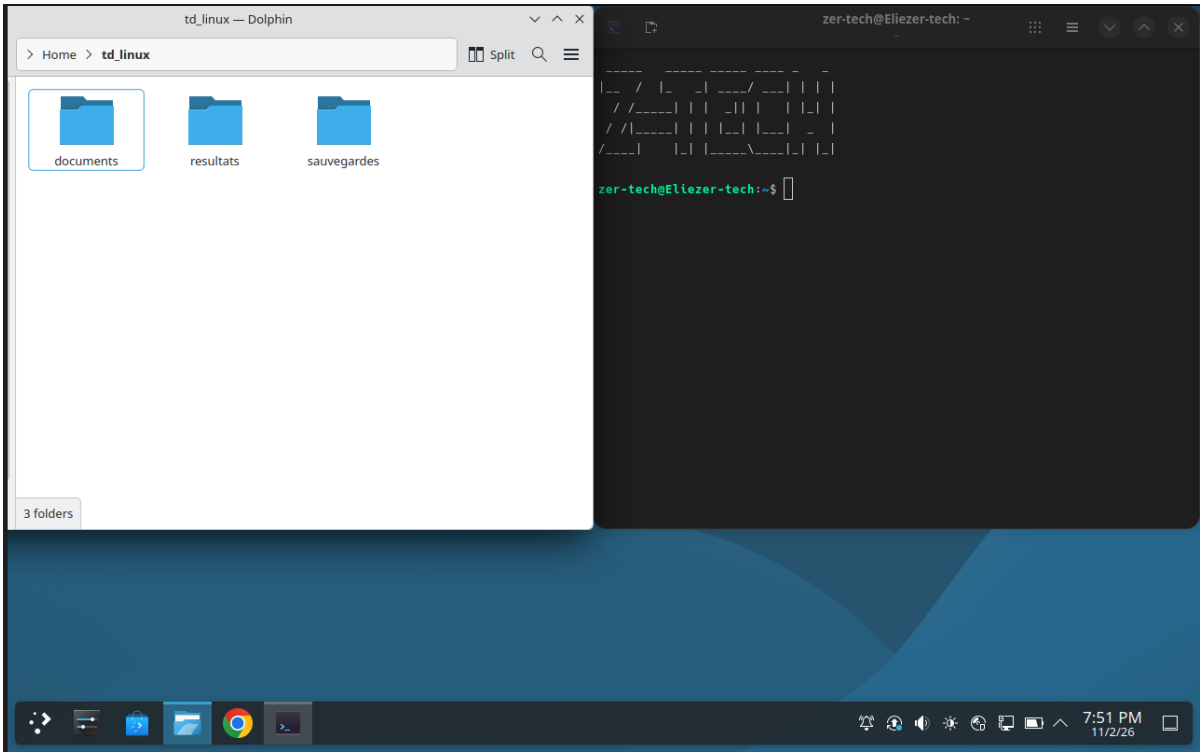

Nov 2 19:13
zer-tech@Eliezer-tech: ~ — sudo apt install kde-plasma-desktop
/dev/sda2 29G 12G 16G 42% /
zer-tech@Eliezer-tech:~$ sudo apt install kde-plasma-desktop
[sudo: authenticate] Password:
The following package was automatically installed and is no longer required:
grub-pc-bin
Use 'sudo apt autoremove' to remove it.
Installing:
kde-plasma-desktop
Installing dependencies:
aha
appmenu-gtk-module-common
appmenu-gtk3-module
apt-config-icons-large
apt-config-icons-large-hidpi
baloo6
bluedevil
breeze
breeze-cursor-theme
breeze-gtk-theme
breeze-wallpaper
budgie-sddm-theme
bup
bup-doc
catdoc
clinfo
libkf5iconthemes5
libkf5kirigami2-5
libkf5style5
libkf5waylandclient5
libkf5widgetsaddons-data
libkf5widgetsaddons5
libkf5windowsystem-data
libkf5windowsystem5
libkf6archive-data
libkf6archive6
libkf6attica6
libkf6auth-data
libkf6authcore6
libkf6baloo6
libkf6balooengine6
libkf6baloowidgets6
libkpmcore13
libkquickcontrolsprivate0
libkscreen-bin
libkscreen-data
libkscreenlocker6
libksysguard-bin
libksysguard-data
libksysguardformatter2
libksysguardsensorfaces2
libksysguardsensors2
libksysguardsystemstats2
libkwaylandclient6
libkwin6
libkworkspace6-6
liblayershellqtinterface6
liblilv-0-0
libxcb-record0
libxcb-render-util0
libxcb-xinerama0
libxcb-xinput0
libxcb-xkb1
libxdgutilsbasedir1.0.1
libxdgutilsdesktopentry1.0.1
libxkbcommon-x11-0
libxml2-utils
libxmlsec1-1
libxmlsec1-openssl1
libxss1
libxvidcore4
libxvmc1
libyuv0
libzing2

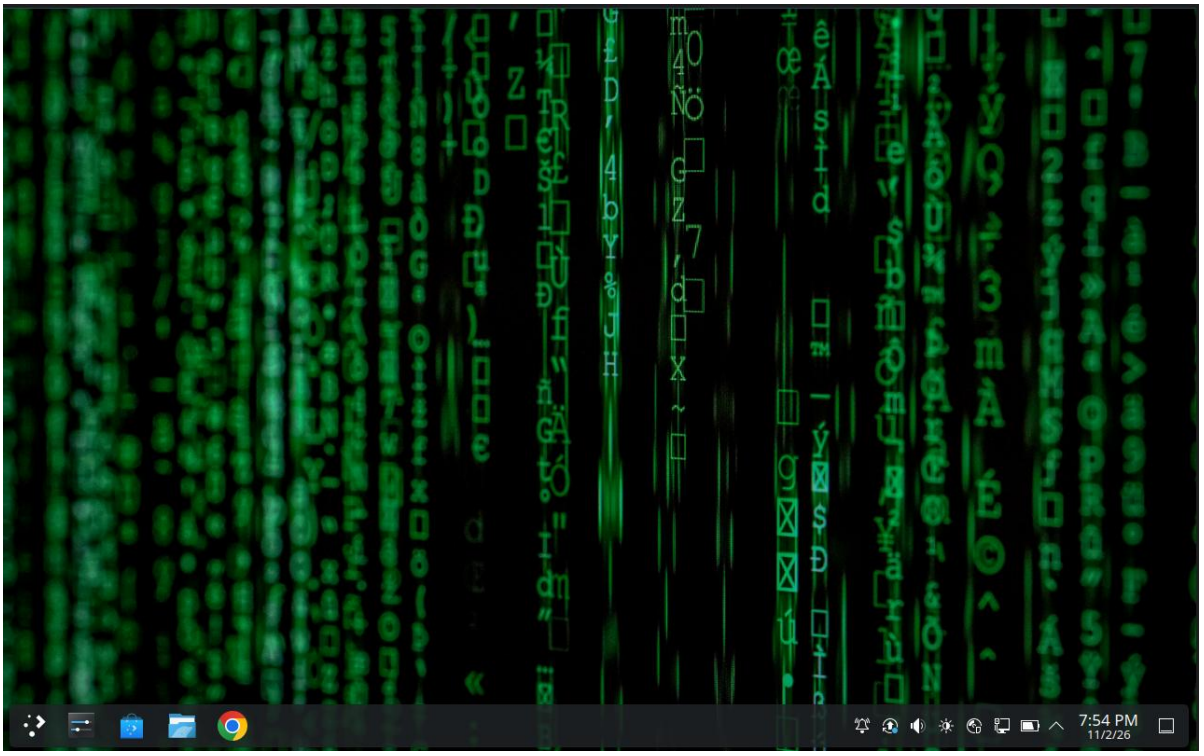

Nov 2 19:23
zer-tech@Eliezer-tech: ~ — sudo apt install xfce4
zer-tech@Eliezer-tech:~$ sudo apt install xfce4
The following package was automatically installed and is no longer required:
grub-pc-bin
Use 'sudo apt autoremove' to remove it.
Installing:
xfce4
Installing dependencies:
desktop-base
elementary-xfce-icon-theme
exo-utils
fonts-quicksand
greybird-gtk-theme
libcairomm-1.16-1
libdbus-glib-1-2
libexo-2-0
libexo-common
libgarcon-1-0
libgarcon-common
libgarcon-gtk3-1-0
libglibmm-2.68-1t64
libglu1-mesa
libgtk-layer-shell0
libgtkmm-4.0-0
libkeybinder-3.0-0
liblightdm-gobject-1-0
libpangomm-2.48-1t64
libsigc++-3.0-0
libthunarx-3-0
libtumbler-1-0t64
libxfce4panel-2.0-4
libxfce4ui-2-0
libxfce4ui-common
libxfce4ui-utils
libxfce4util-bin
libxfce4util-common
libxfce4util7
libxfce4windowing-0-0
libxfce4windowing-common
libxfconf-0-3
libxklavier16
libxpresent1
light-locker
light-locker-settings
lightdm
lightdm-gtk-greeter
mate-polkit
mate-polkit-common
pavucontrol
tango-icon-theme
thunar
thunar-data
thunar-volman
tumbler
tumbler-common
x11-apps
x11-session-utils
xbitmaps
xfce4-appfinder
xfce4-helpers
xfce4-notifyd
xfce4-panel
xfce4-pulseaudio-plugin
xfce4-session
xfce4-settings
xfconf
xfdesktop4
xfdesktop4-data
xfonts-scalable
xfwm4
xiccd
xinit
xinput
xorg
xorg-docs-core
Suggested packages:
devhelp
gnome-icon-theme
thunar-media-tags-plugin
xfce4-goodies
fortune-mod
xorg-docs
xfonts-75dpi
xserver-xephyr
thunar-archive-plugin
tumbler-plugins-extra
xfce4-power-manager
xfconf-gsettings-backend
xfonts-100dpi
x11-xfs-utils
Summary:

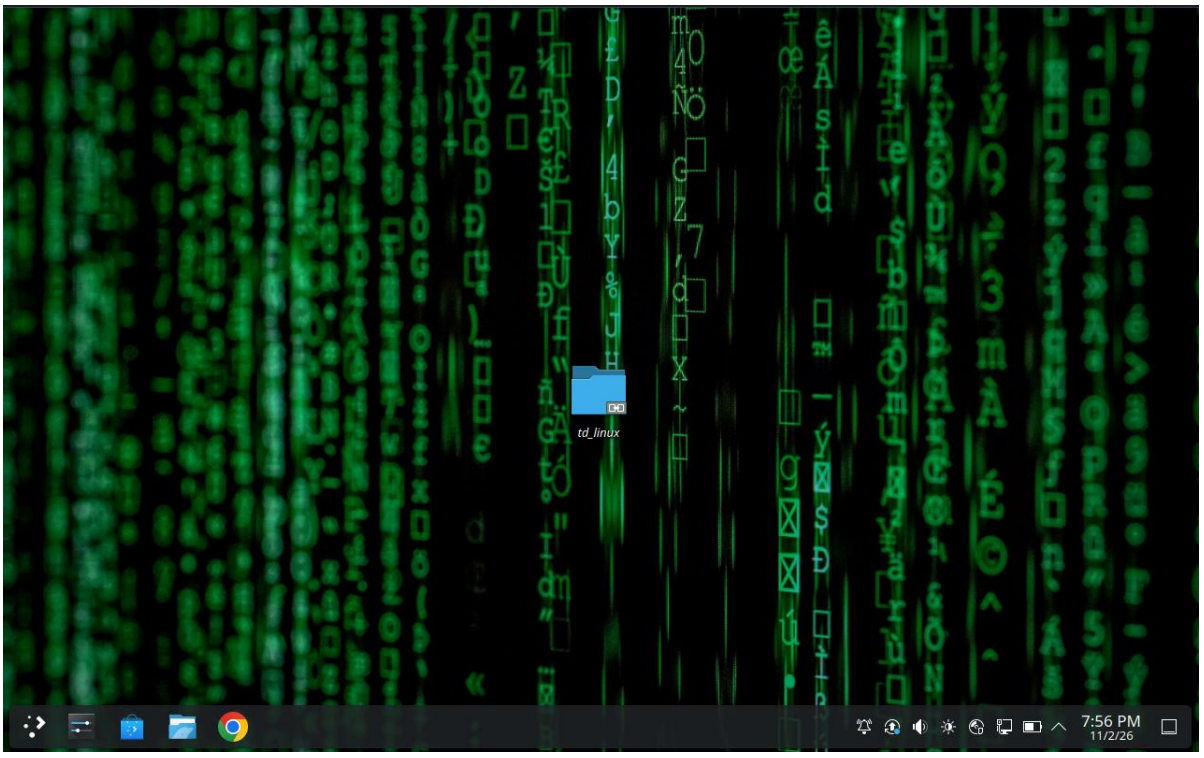

zer-tech@Eliezer-tech: ~ — sudo apt install xfce4
Package configuration
Configuring lightdm
A display manager is a program that provides graphical login capabilities for the X Window System.
Only one display manager can manage a given X server, but multiple display manager packages are installed. Please select which display manager should run by default.
Multiple display managers can run simultaneously if they are configured to manage different servers; to achieve this, configure the display managers accordingly, edit each of their init scripts in /etc/init.d, and disable the check for a default display manager.
Default display manager:
gdm3
lightdm
sddm
<Ok>

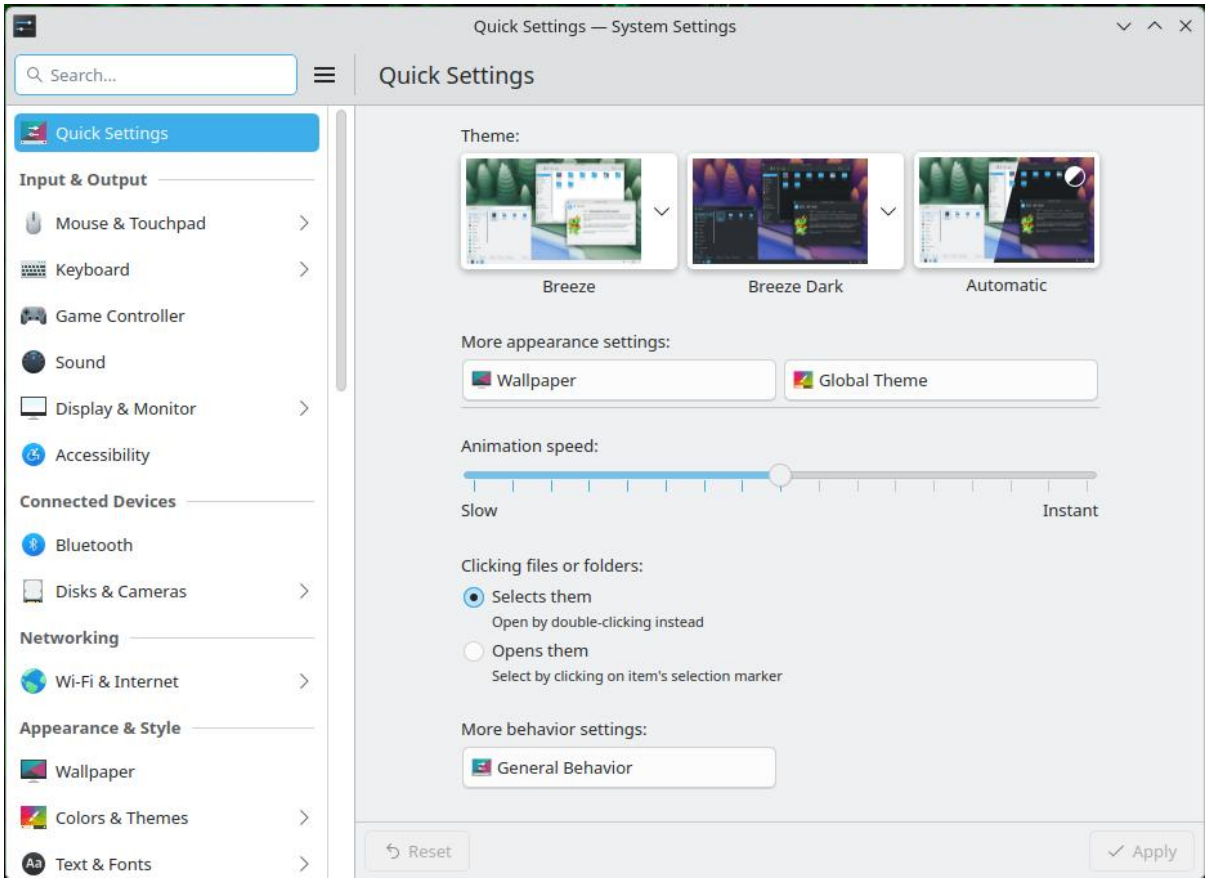

Nov 2 19:30
zer-tech
Password
Session
GNOME
GNOME Classic
Default Xsession
Plasma (Wayland)
Ubuntu
Xfce Session
Xfce Session (Wayland)
Ubuntu

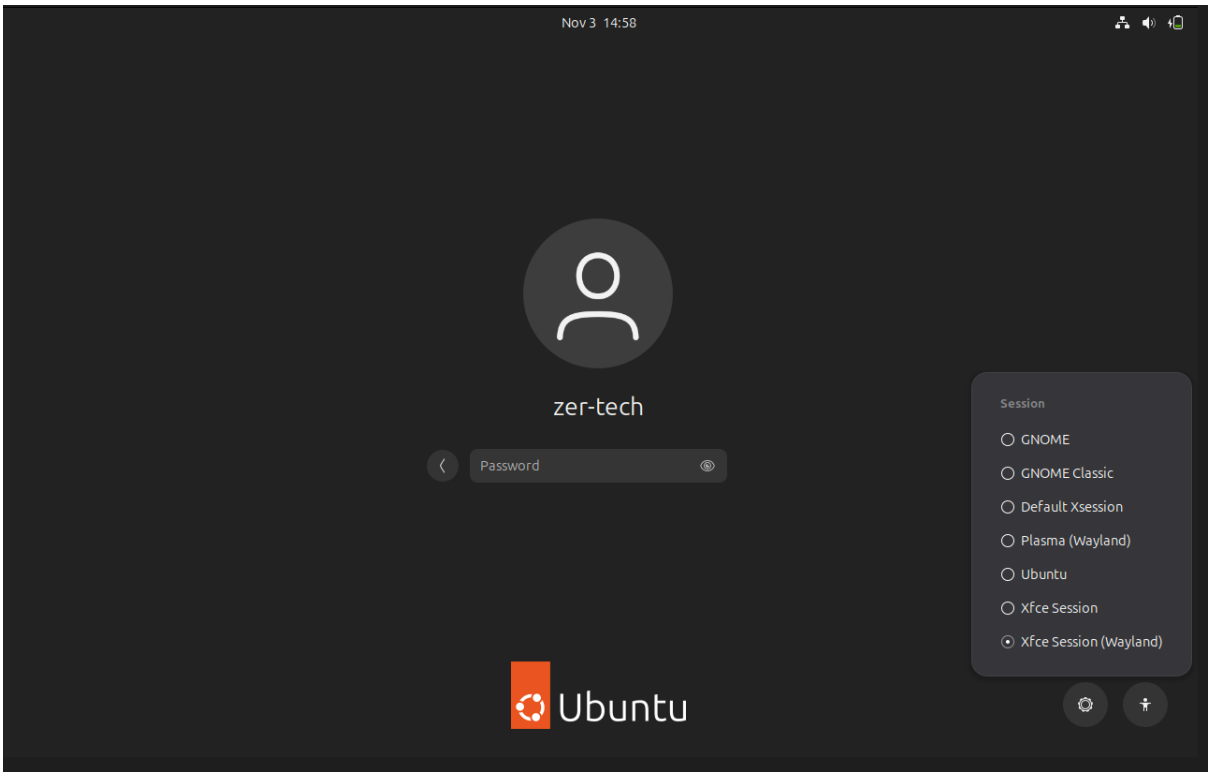

Ubuntu
Sorry, Ubuntu 26.04 has experienced an internal error.
If you notice further problems, try restarting the computer.
Show Details
Continue
snapd-desktop-integration
Some required themes are missing.
Would you like to install them now?
Yes
No
7:31 PM
11/2/26

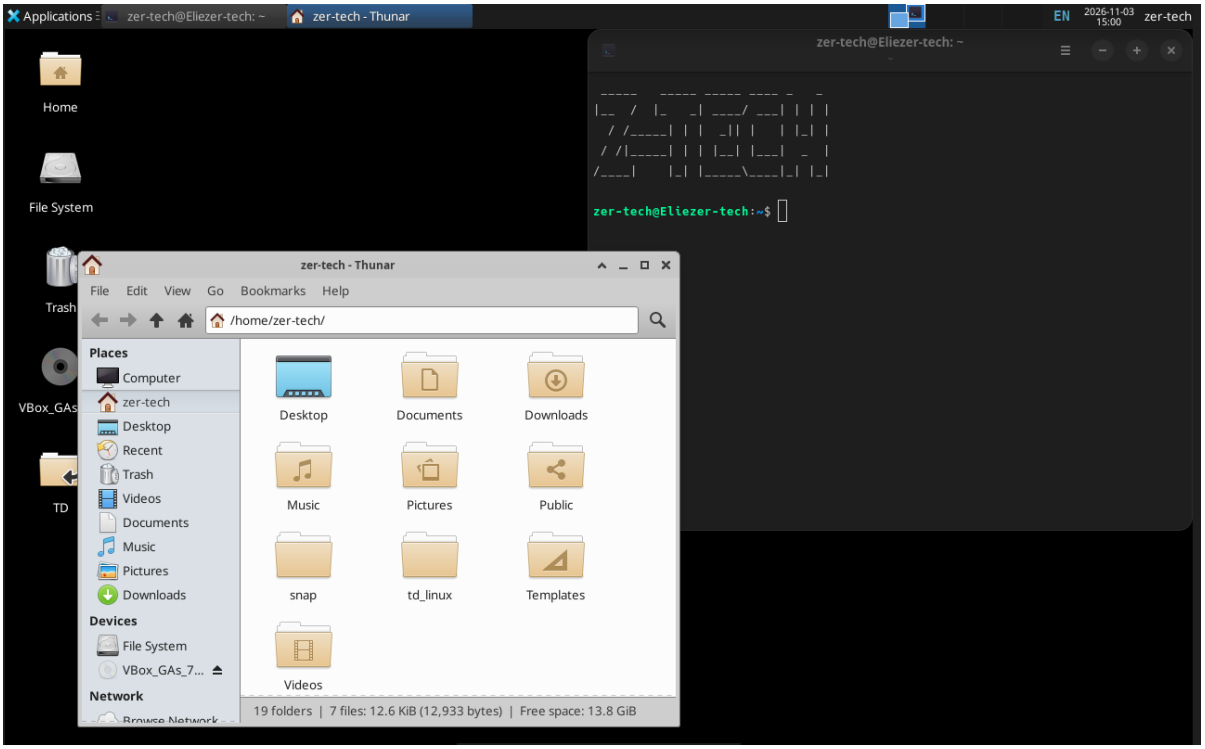

td_linux — Dolphin
Home
td_linux
Split
documents
resultats
sauvegardes
3 folders
zer-tech@Eliezer-tech: ~
zer-tech@Eliezer-tech:~$
7:51 PM
11/2/26

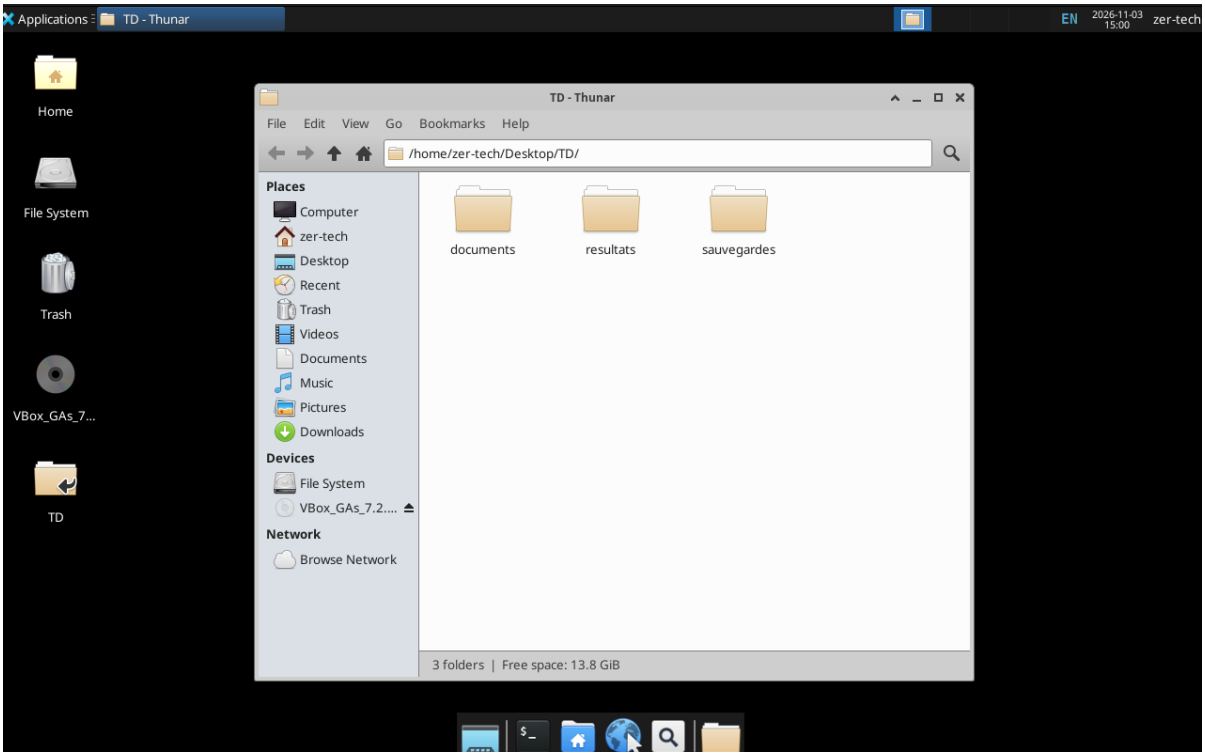

7:54 PM
11/2/26


td_linux
7:56 PM
11/2/26

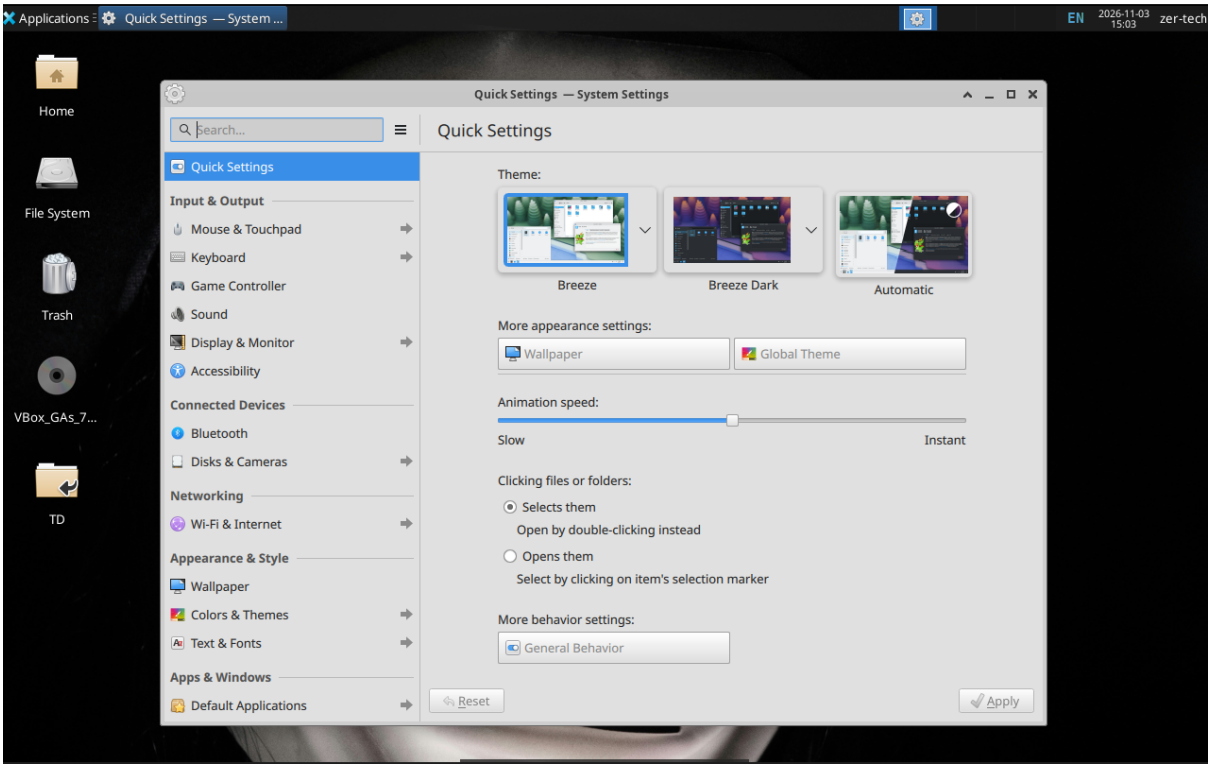

Quick Settings — System Settings
Search...
Quick Settings
Quick Settings
Input & Output
Mouse & Touchpad
Keyboard
Game Controller
Sound
Display & Monitor
Accessibility
Connected Devices
Bluetooth
Disks & Cameras
Networking
Wi-Fi & Internet
Appearance & Style
Wallpaper
Colors & Themes
Text & Fonts
Theme:
Breeze
Breeze Dark
Automatic
More appearance settings:
Wallpaper
Global Theme
Animation speed:
Slow
Instant
Clicking files or folders:
Selects them
Open by double-clicking instead
Opens them
Select by clicking on item's selection marker
More behavior settings:
General Behavior
Reset
Apply


Nov 3 14:58
zer-tech
Password
Session
GNOME
GNOME Classic
Default Xsession
Plasma (Wayland)
Ubuntu
Xfce Session
Xfce Session (Wayland)
Ubuntu


Applications
zer-tech@Eliezer-tech: ~
zer-tech - Thunar
EN
2026-11-03
15:00
zer-tech
Home
File System
Trash
VBox_GAs
TD
zer-tech@Eliezer-tech:~$
File Edit View Go Bookmarks Help
/home/zer-tech/
Places
Computer
zer-tech
Desktop
Recent
Trash
Videos
Documents
Music
Pictures
Downloads
Devices
File System
Network
Desktop
Documents
Downloads
Music
Pictures
Public
snap
td_linux
Templates
Videos
19 folders | 7 files: 12.6 KiB (12,933 bytes) | Free space: 13.8 GiB


Applications
TD - Thunar
EN
2026-11-03
15:00
zer-tech
Home
File System
Trash
VBox_GAs_7...
TD
File Edit View Go Bookmarks Help
/home/zer-tech/Desktop/TD/
Places
Computer
zer-tech
Desktop
Recent
Trash
Videos
Documents
Music
Pictures
Downloads
Devices
File System
VBox_GAs_7.2....
Network
Browse Network
documents
resultats
sauvegardes
3 folders | Free space: 13.8 GiB


Applications
EN
2026-11-03
15:01
zer-tech
Home
File System
Trash
VBox_GAs_7...
TD


Applications
Quick Settings — System ...
EN
2026-11-03
15:03
zer-tech
Home
File System
Trash
VBox_GAs_7...
TD
Quick Settings — System Settings
Search...
Quick Settings
Quick Settings
Input & Output
Mouse & Touchpad
Keyboard
Game Controller
Sound
Display & Monitor
Accessibility
Connected Devices
Bluetooth
Disks & Cameras
Networking
Wi-Fi & Internet
Appearance & Style
Wallpaper
Colors & Themes
Text & Fonts
Apps & Windows
Default Applications
Theme:
Breeze
Breeze Dark
Automatic
More appearance settings:
Wallpaper
Global Theme
Animation speed:
Slow
Instant
Clicking files or folders:
Selects them
Open by double-clicking instead
Opens them
Select by clicking on item's selection marker
More behavior settings:
General Behavior
Reset
Apply

Critère	KDE Plasma	XFCE
Terminal Utilisé	konsole	xfce4-terminal
Gestionnaire de fichiers	Dolphin	Thunar
Accès aux paramètres	systemesetting	xfce-4-settings-manager
personnalisation	Tout est modifiable depuis l'interface sans avoir besoin de faire recours à un outil tiers	personnalisation limitée, simple rapide consomme peu de ressources
mémoire disponible	1.7gi	2.2gi
choix pour un poste donné	poste de travail avec des matériels récents, avec confort visuel au max	poste de travail pour des matériels bas de gamme ou ancien, pour limiter la consommation de RAM

Conclusion

En conclusion, nous appris à manipuler le terminal sur linux et effectuer certaines tâches comme créer et manipuler des dossiers, créer des archives sans avoir besoin d'utiliser l'interface graphique. Ce qui fait la force et la maniabilité extrême de linux.

L'autre point important de ce TD c'est aussi l'initialisation aux packets, nous avons appris à simuler leur installation et suppression via *apt -s install*. La commande *sudo apt install*, collectionne les dépendances des packets et les installe d'un coup.

Difficile de passer a cote de l'étape visant à configurer les deux espaces de travail KDE plasma et XFCE, pendant l'installation nous avons dû changer l'affichage par défaut au dépend de GDN3 pour pouvoir utilisé les différents affichages.

Les difficultés que nous avons rencontré lors de ce travail était principalement dans l'exercice 2 dans laquelle nous avons créé le fichier services.txt via la commande : *cat > documents/services.txt <<'EOF' web;actif ssh;actif impression;inactif sauvegarde;actif web;actif EOF*. Mais malheureusement nous avons dû opter pour une autre option vue celle-ci était défaillante.
Nous avons aussi rencontré quelques difficultés mineurs par rapport à l'affichage de KDE plasma. et les packets cmatrix, neofetch,ncal et cawsay était absent nous les avons installé .

En somme le travail est réussi, et nous sommes en mesure de travailler sur différents espaces de travail soit Ubuntu, DKE plasma ou XFCE.